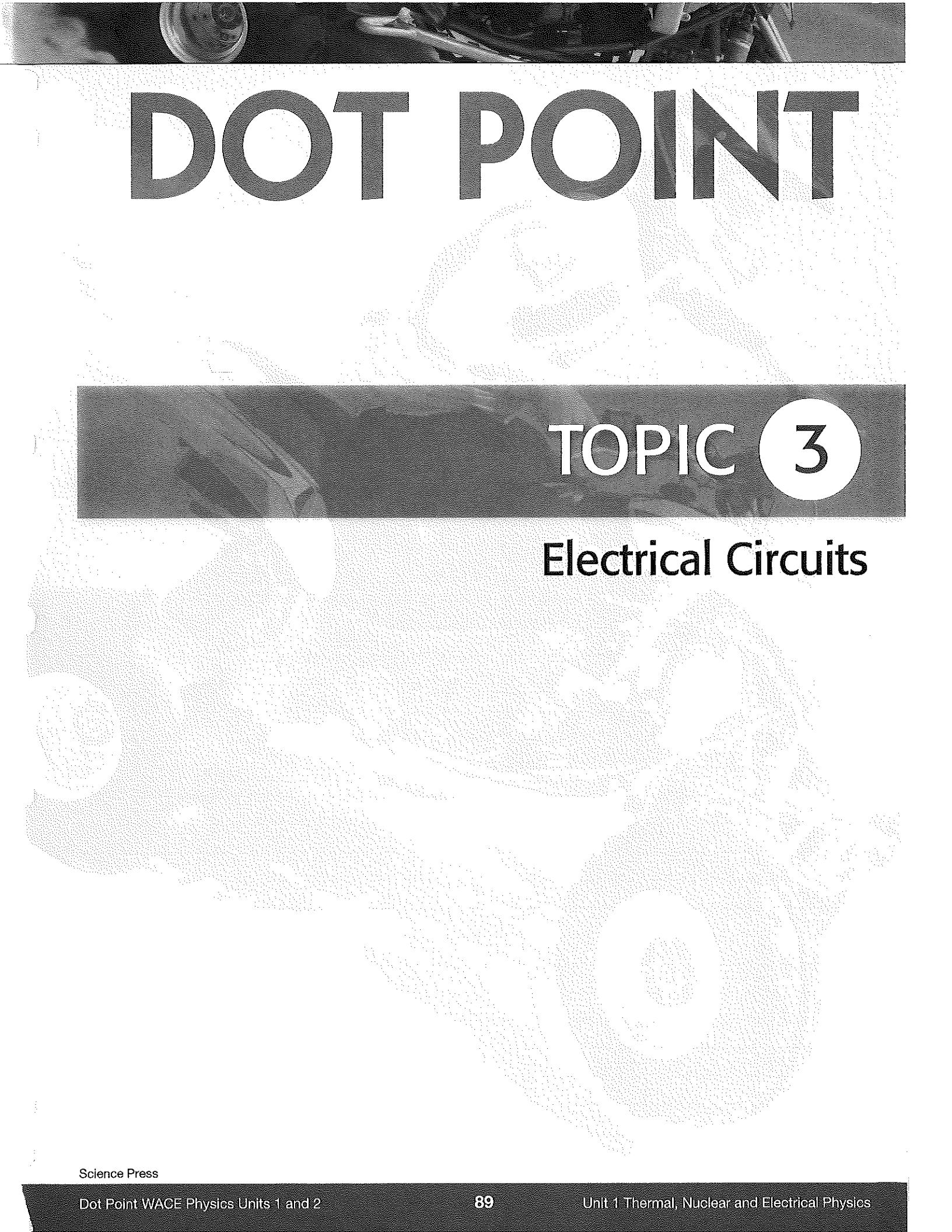




DOT POINT

TOPIC 3

Electrical Circuits



Notes

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3.1 Electric current and charge

3.1.1 The charge model for electric current.

3.1.1.1

- (a) What is the source of all electric charges?
- (b) How does a neutral object become positively charged?
- (c) How does a neutral object become negatively charged?
- (d) Explain why charges refer to excesses or deficiencies of electrons rather than protons.
.....
.....

- (e) What is the unit we use to measure electric charge and its symbol?
- (f) Define the unit of electric charge.
.....

- (g) An object has a deficiency of 6.25×10^{15} electrons. What charge does it have?

3.1.1.2

- (a) Define electric current in terms of electron flow.
.....
- (b) What is the unit for electric current, its symbol and the unit we use to measure it?
- (c) Outline the difference between electron flow and the flow of conventional current in a circuit.
.....
.....

- (d) If current electricity is a flow of electrons, why do we have the idea of 'conventional' current at all?
.....
.....
.....

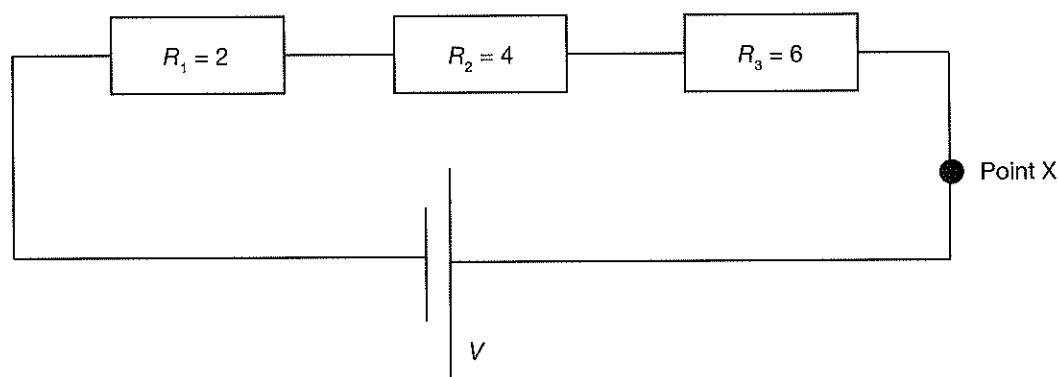
3.1.1.3

- (a) A current of 2.4 A flows through a conductor for 8 s. How many electrons pass each point in the conductor in this time?
.....
- (b) If 200 mA of current flows in a conductor, how long would it take for 3.0 coulombs of electrons to pass a point in the wire?
.....

3.1.1.4

- (a) 1.2×10^{19} electrons flow past a point in a circuit each second. What is the current in the circuit?
-
- (b) 5.6×10^{17} electrons flow past a point in a circuit each minute. What is the current in the circuit?
-
- (c) How many electrons flow past a point in a circuit in one second if the current in the circuit is 0.125 A?
-
- (d) How many electrons flow past a point in a circuit in one second if the current in the circuit is 3.5×10^{-9} A?
-

3.1.1.5 Consider the following electrical circuit. The relative sizes of the resistors are given.



- (a) How much current will flow through R_1 compared to that flowing through X?
-
- (b) How much current will flow through R_1 compared to that flowing through R_2 ?
-
- (c) How much current will flow through R_2 compared to that flowing through R_3 ?
-
- (d) Compare the number of electrons per second flowing through R_1 and R_2 and R_3 .
-

3.2 Charge carriers and electrical potential difference.

3.2.1 Electrical potential difference 1.

3.2.1.1 Match the sentence halves to form complete sentences. Write your answers in the rows provided and choose from the alternate second halves given under the table.

	Completed sentences
(a)	Energy is required to
(b)	When charges are separated
(c)	Whenever work is done to separate objects a potential
(d)	This is how an electrical potential
(e)	Potential difference is defined as the work done per unit positive test charge in moving the charge from
(f)	The direction of the potential difference at P is given by the direction of the force which would
(g)	If a charge is positive, then work needs to be done on it
(h)	If a charge is negative, then work will need to be done on it
(i)	If a charge is positive, then work is done on it by
(j)	If a charge is negative, then work is done on it by
(k)	In general, electrical potential difference is directed away from
(l)	The direction of electrical potential difference is the direction in which a positive
(m)	The potential energy difference set up when charges are separated from each other can be used as the

Second halves of sentences (not in correct order)

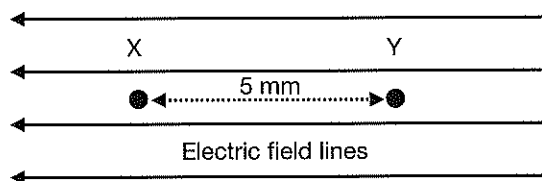
- (A) one point in an electric field to another point in the field.
- (B) to move it in the direction of the field.
- (C) separate positive and negative charges.
- (D) energy difference is set up between them.
- (E) act on a positive test charge placed at P.
- (F) work is done.
- (G) the field to move it in the direction of the field.

- (H) to move it against the field.
- (I) positive charge and towards negative charge.
- (J) energy difference is produced.
- (K) test charge would move if it was placed in the potential field.
- (L) driving force to produce an electric current in a circuit.
- (M) the field to move it in the opposite direction of the field.

- 3.2.1.2** 7.2×10^{-2} J of work is done moving a charge of 25 mC from X to Y. What is the potential difference between X and Y?
- (A) 2.88×10^{-3} V
(B) 1.8×10^{-3} V
(C) 0.35 V
(D) 2.88 V
- 3.2.1.3** The work done by an external force in moving a 5.0×10^{-6} C charge from X to Y is 6×10^{-2} J. If the charge was initially at rest and had 1.5×10^{-3} J of kinetic energy at point Y, what is the electrical potential difference between X and Y?
- (A) 300 V
(B) 900 V
(C) 11 700 V
(D) 12 000 V
- 3.2.1.4** How much work is done when a total charge of 4.0 mC flows through an electrical device if the potential difference across the device is 240 V?
- (A) 0.96 J
(B) 60 J
(C) 960 J
(D) 60 000 J
- 3.2.1.5** What is the voltage of a battery that supplies 3 J of energy to every 0.25 C of charge passing through it?
- (A) 0.08 V
(B) 0.75 V
(C) 4.25 V
(D) 12 V
- 3.2.1.6** What is the voltage of a calculator battery which provides each electron passing through it with 3.6×10^{-19} J of electrical potential energy?
- (A) 0.44 V
(B) 2.25 V
(C) 5.76 V
(D) 22.5 V
- 3.2.1.7** What work is done in moving 4.5 C of charge through an electrical device which is plugged into a 110 V power source?
- (A) 2.06 J
(B) 24.4 J
(C) 45 J
(D) 494 J

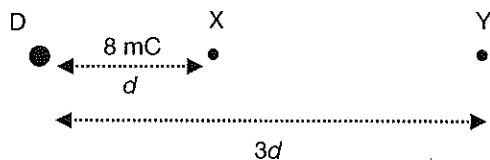
Use the following information to answer the next THREE questions.

Points X and Y are 5 mm apart in an electric field. When a charge of +6.0 nC is moved from X to Y, the field does 3×10^{-6} J of work.



- 3.2.1.8** What is the magnitude of the potential difference between X and Y?
- (A) 1.8×10^{-14} V
(B) 2.0×10^{-3} V
(C) 500 V
(D) 2000 V
- 3.2.1.9** If it takes 5.0 ms for the charge to move from X to Y, what was the equivalent flow of current?
- (A) 1.2×10^{-6} A
(B) 12 mA
(C) 12 μ A
(D) 12 nA
- 3.2.1.10** Which of the following statements is correct?
- (A) Potential difference is directed from X to Y and work is done by the field in moving the charge.
(B) Potential difference is directed from Y to X and work is done by the field in moving the charge.
(C) Potential difference is directed from X to Y and work is done by an external force to move the charge.
(D) Potential difference is directed from Y to X and work is done by an external force in moving the charge.
- 3.2.1.11** You charge a plastic ruler positively by rubbing the bottom side of it on your sleeve. In doing this you set up a potential difference of 120 V between the two sides.
- How much work would have to be done to move a charge of 3.2 μ C, from one side of the ruler to the other?
- (A) 2.67×10^{-8} J
(B) 3.84×10^{-4} J
(C) 0.38 J
(D) 3.75×10^7 J

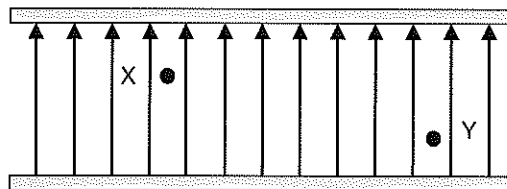
- 3.2.1.12** Three points X, Y and Z are distances d and $3d$ apart in a uniform electric field. If it takes $4.8 \times 10^{-6} \text{ J}$ to move a charge of $1.6 \times 10^{-8} \text{ C}$ from X to Y, what is the potential difference between Y and Z?



- (A) 1500 V
- (B) 3000 V
- (C) 6000 V
- (D) 9000 V

Use the following information to answer the next FIVE questions.

The diagram shows electric field lines between two parallel plates and two positions X and Y, between them. A charge of $-6.4 \times 10^{-18} \text{ C}$ is placed at point X. It is then moved to point Y.



- 3.2.1.13** Which statement is correct?
- (A) The top plate is positive and X is at a higher potential than Y.
 - (B) The top plate is positive and Y is at a higher potential than X.
 - (C) The bottom plate is positive and X is at a higher potential than Y.
 - (D) The bottom plate is positive and Y is at a higher potential than X.

- 3.2.1.14** Which statement about the force on the charge when it is at X compared to when it is at Y is correct?

- (A) The force on the charge is equal at X and Y and directed towards the bottom plate.
- (B) The force on the charge is equal at X and Y and directed towards the top plate.
- (C) The force on the charge is greater at X than at Y and directed towards the bottom plate.
- (D) The force on the charge is less at X than at Y and directed towards the top plate.

- 3.2.1.15** Which statement about the movement of the charge from X to Y is correct?

- (A) Work is done on the charge by the electric field and its kinetic energy increases.
- (B) Work is done on the charge by an external source and its potential energy increases.
- (C) No work was done on the charge because the field between parallel plates is constant magnitude.
- (D) No work was done on the charge and its energy does not change.

- 3.2.1.16** Suppose the charge had been a positive charge instead of a negative charge. What would have been your answer to the previous question?

- (A) A
- (B) B
- (C) C
- (D) D

- 3.2.1.17** An alpha particle (He^{2+}) gains $4.8 \times 10^{-18} \text{ J}$ of kinetic energy in moving freely from X to Y. What is the potential difference between points X and Y?

- (A) 15 V from X to Y.
- (B) 15 V from Y to X.
- (C) 30 V from X to Y.
- (D) 30 V from Y to X.

3.3 Resistance for ohmic and non-ohmic components.

3.3.1 Electrical potential difference 2.

3.3.1.1

- (a) Define potential difference between two points in a circuit in terms of the current flowing and the resistance of the conductor.

.....

.....

- (b) Write down the equation we get for potential difference from this definition.

.....

- (c) What unit could we use for potential difference based on this definition?

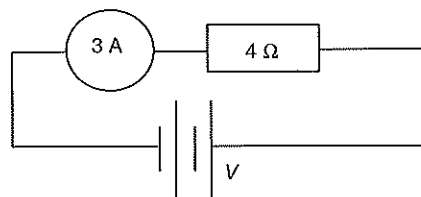
.....

- (d) What is the relationship between this unit and the volt?

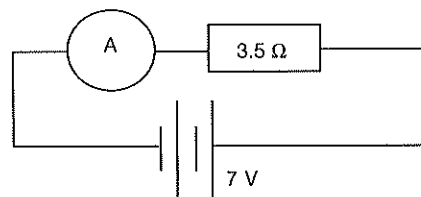
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3.3.1.2

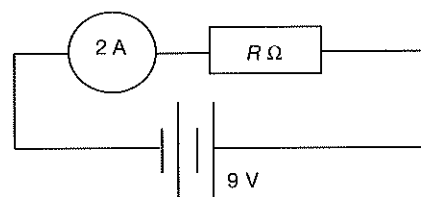
- (a) A current of 3 amperes flows through a four ohm resistor as shown in the simple circuit diagram. What would be the potential difference across the resistor?



- (b) A potential difference of 7 V is applied across the ends of a 3.5 Ω resistor as shown in the simple circuit diagram. What current will flow through this resistor?

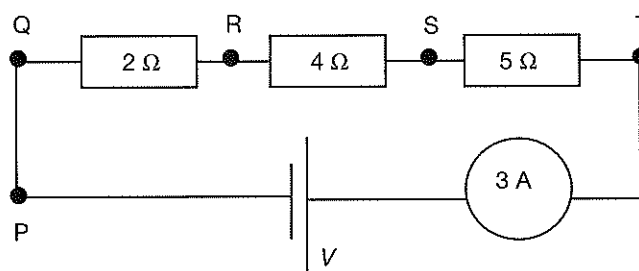


- (c) A potential difference of 9 V is applied across the ends of a resistor as shown in the simple circuit diagram causing a current of 2 A to flow. What is the value of the resistance of this resistor?

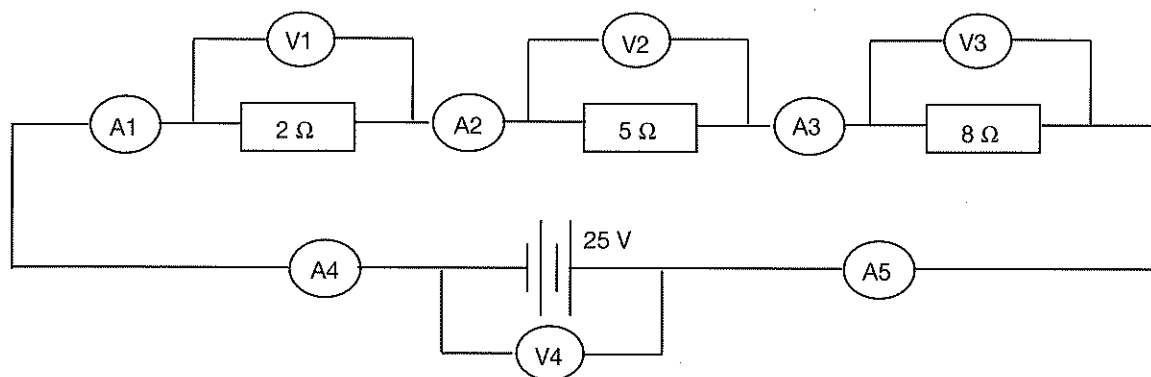


- 3.3.1.3** Consider the electric circuit shown. Using your definition of potential difference above, determine the potential difference across points:

- (a) Q and R.
- (b) R and S.
- (c) R and T.
- (d) Q and T.
- (e) P and Q.



- 3.3.1.4** Students set up the circuit shown to study resistors in series. Their results are shown in the table. All voltages were measured in volts and all currents in amperes.



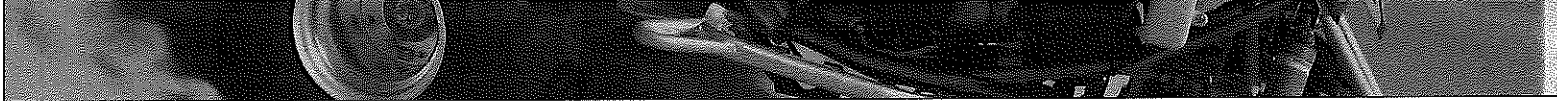
- (a) Complete the table by calculating all averages.

Run	V1	V2	V3	V4	A1	A2	A3	A4	A5
1	3.3	8.5	13.5	25.0	1.8	1.6	1.8	1.7	1.7
2	3.4	8.2	13.4	25.3	1.6	1.7	1.5	1.7	1.8
3	3.3	8.3	13.3	24.7	1.7	1.7	1.7	1.6	1.5
Average									

- (b) Use the results to determine the relationship between the current flowing through different resistors connected in series and the current flowing in the circuit. Include an equation.
- (c) Determine the potential differences across resistors connected in series and the potential difference of the power source. Include an equation.
- (d) Find the relationship between the total resistance of the circuit and the individual resistors. Include an equation.
- (e) Determine the potential difference across resistors connected in series and the value of the resistor. Include an equation.

- (f) Find the electrical potential at the position of each ammeter.

Position of ammeter	Potential at this position (V)
A1	
A2	
A3	
A4	
A5	



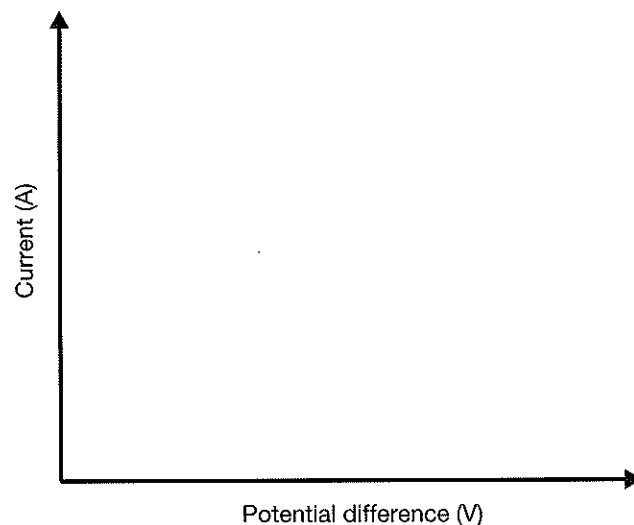
3.3.2 The nature of electrical resistance.

3.3.2.1

- (a) What is electrical resistance?
-
- (b) What do we use resistors for in electrical circuits?
-
- (c) What causes electrical resistance?
-
- (d) State the unit we use to measure electrical resistance and give the symbol we use for it.
- (e) Define this unit.
-
- (f) Give the equation we use to calculate electrical resistance.
-

3.3.2.2

- (a) Imagine a 20 m length of a constant resistance conductor at 20°C. Draw a graph on the axes provided to show the relationship between the potential difference across the ends of the conductor and the current flowing through it. Label this graph X.
- (b) Draw a similar graph for the relationship if the conductor was 40 m long. Label this graph Y.
- (c) Explain any differences in the graph you drew for Y compared to the one you drew in X.
-
-
-
-
- (d) Draw a similar graph for the same conductor you imagined in (a) operating at 40°C. Label this graph Z. Assume that the conductor remains ohmic.
- (e) Explain any differences in graph Z compared to graph X.
-
- (f) You want to do an experiment to study the relationship between the resistance of a conductor and its length. Briefly describe, listing equipment, variables, controls and method, how you would do this experiment.
-
-
-
-



3.3.2.3

- (a) Explain why temperature affects the resistance of a conductor.
-
-
- (b) Explain why the length of the conductor affects its resistance.
-
-
- (c) Explain why the composition of a conductor affects its resistance.
-
-
- (d) Explain why the cross-sectional area of a conductor affects its resistance.
-
-
-

3.3.2.4 Complete the table by writing the names of electrical devices in column 1 and a useful output form of energy derived from them in column 3. Note that each output energy must be a different type.

Electrical device	Input energy	Useful output energy
	Electrical	
	Electrical	
	Electrical	
	Electrical	
	Electrical	

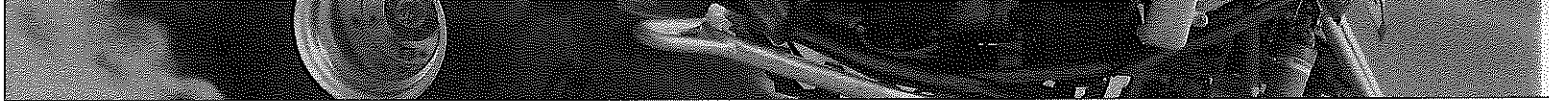
3.3.2.5 Household circuits are often wired with two different widths of wires. These include 10 gauge and 18 gauge. The 10 gauge wire has a diameter of 1/10 inch while the 18 gauge wire has a diameter of 1/18 inch. A 15 ampere circuit used for power points should be wired using 10 gauge wire and a 10 ampere circuit used for lighting and fan circuits could be wired using 18 gauge wire. Explain the physics behind such an electrical code.

.....

.....

.....

.....

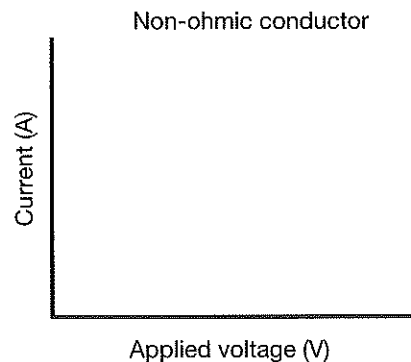
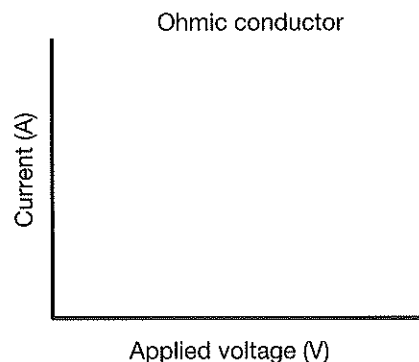


3.3.3 Ohmic and non-ohmic conductors.

3.3.3.1

(a) Distinguish between an ohmic and a non-ohmic conductor.
.....
.....

(b) Sketch graphs on the axes below to show the relationship between the current flowing in ohmic and non-ohmic conductors and the applied potential difference.



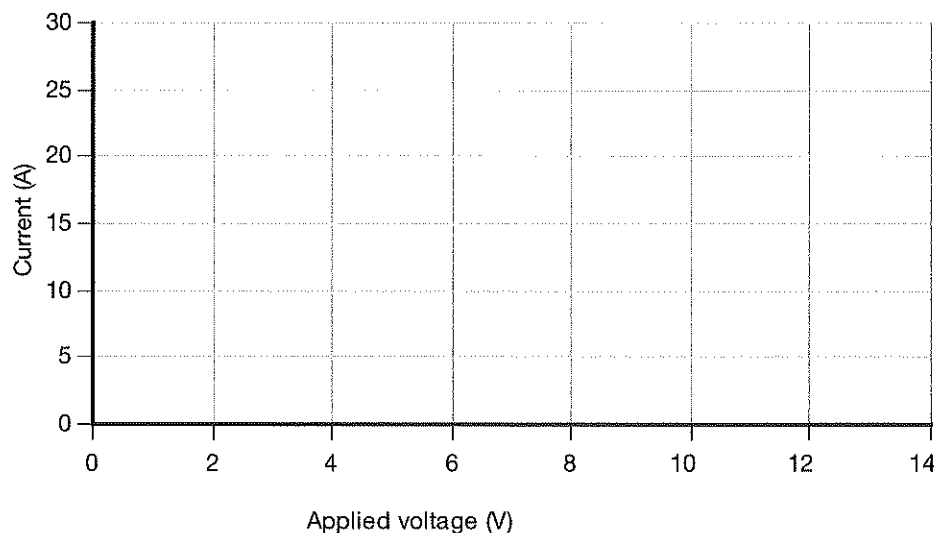
(c) Comment on the magnitude of the resistance of an ohmic conductor as the voltage across it increases.
.....

(d) Comment on the magnitude of the resistance of a non-ohmic conductor as the voltage across it increases. Explain why this happens.
.....

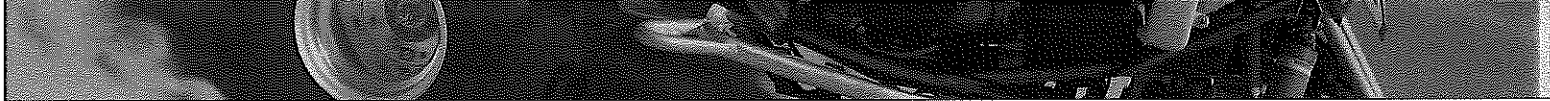
3.3.3.2 The table shows the results of an experiment done by a group of students who were studying the relationship between applied potential difference across a conductor and the current produced.

Applied potential difference (V)	Current produced in conductor X (A)	Current produced in conductor Y (A)
0	0	0
2	3.9	4.1
4	8.2	6.9
6	11.7	9.6
8	16.9	11.4
10	20.3	13.7
12	23.8	14.9
14	28.1	15.8

- (a) Plot the results of this experiment on the axes provided and complete the graphs.



- (b) Comment on the shape of the graph for each of the two conductors.
.....
.....
- (c) Explain the reasoning for these results.
.....
.....
- (d) What is the value of the resistance of conductor X?
- (e) What is the best way to determine this value? Explain your answer.
.....
.....
- (f) What is the value of the resistance of conductor Y when a potential of 10 V is applied?
- (g) With specific reference to these results, explain how using a graph rather than calculating individual values of the resistances of the conductors improves the reliability of the experiment.
.....
.....
.....
- (h) How might this experiment be improved?
.....



3.4 Electrical power and energy transformations.

3.4.1 Electrical power.

3.4.1.1

- (a) Define power in terms of work done.
.....
- (b) Define potential difference in terms of power.
.....
- (c) Write down the equations we get for potential difference from this definition.
.....
- (d) What unit could we use for potential difference based on this definition?
- (e) What is the relationship between this units and the volt?
- (f) Give three alternate units we could use for measuring power.
.....

3.4.1.2

- (a) State the relationship between the voltage of an appliance, the current flowing through it and its power.
.....
- (b) Using this relationship you expressed in (a), and the Ohm's law equation, write three equations we can use for finding the electrical power of an appliance.
.....
- (c) Calculate the number of joules in a kWh.
- (d) Outline why electricity is sold in kWh units rather than joules.
.....

3.4.1.3

- (a) Given that 1 kWh costs about 21 cents, calculate the cost of 1 joule of electricity.
- (b) Explain the difference between electrical energy and electrical power.
.....

3.4.1.4 A toaster is rated 240 V, 800 W. Calculate:

- (a) The current it will draw.
- (b) The resistance of its element.
- (c) How many hours of operation would use 35 kWh.

3.4.1.5 A potential difference of 18 V is applied across the ends of a resistance of $6\ \Omega$.

- (a) Calculate the power developed by the conductor in watts.
- (b) Calculate the energy dissipated (changed into heat and/or light energy) in 15 s.
- (c) Calculate the energy the conductor uses, in kWh, in 30 days.

3.4.1.6 A toaster is rated at 240 V, 800 W.

- (a) Calculate the current it will draw when being used.

.....
.....

- (b) Calculate how much electrical energy it consumes if it takes 2.5 minutes to toast bread.

.....
.....

- (c) If electricity costs 12 cents per kWh, calculate how much it would cost to use the toaster each day for a year if it was used to make two rounds of toast each day.

.....

3.4.1.7 Some globes we use in our homes are marked 240 V 40 W.

- (a) Calculate the current these globes draw.
- (b) Calculate the resistance of their filaments.
- (c) A similar globe is marked 40 W but its voltage reading is blurred. Its resistance is measured to be $22.5\ \Omega$. Calculate its operating voltage.

.....

- (d) Predict what would happen to this globe if it was used in a 240 V circuit.

.....
.....

3.4.1.8 The heating coil of an electric jug consists of 6 m of a thin conductor. It develops 2400 W when connected to 240 V supply.

- (a) Calculate the power it would develop if used overseas in a 110 V circuit.
- (b) Comment on the efficiency of the jug under these conditions.
- (c) Calculate the current the jug would draw in Australia.
- (d) Calculate the current it would draw overseas with a 110 V supply.



3.4.2 Analysing an experiment.

3.4.2.1 A group of students made a heating coil from a piece of resistance wire and used it to heat separate amounts of 100 grams of water in an insulated container.

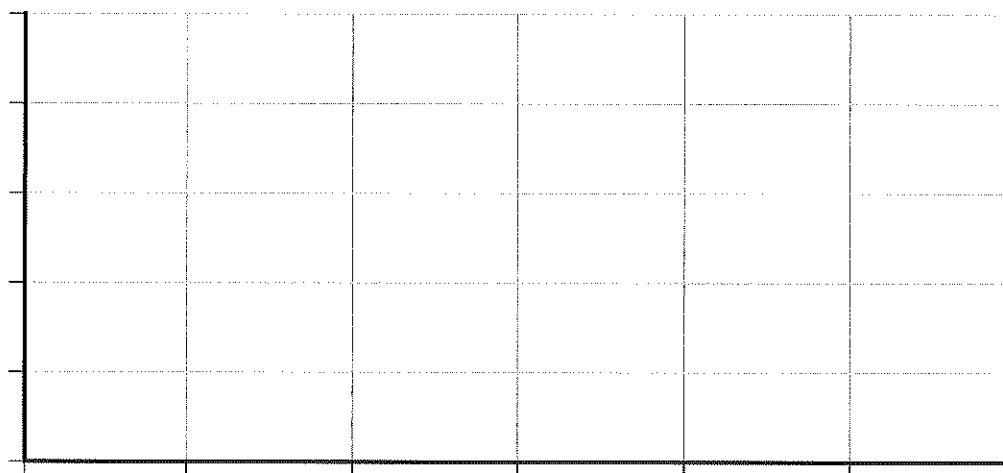
They connected an external, variable resistor into the circuit so that each time the applied potential difference was increased, the resistor could be adjusted to maintain a constant current.

They measured the temperature rise in the water each time using a heat probe which was connected to a data logger, and subsequently downloaded onto a computer.

Their results are shown in the table.

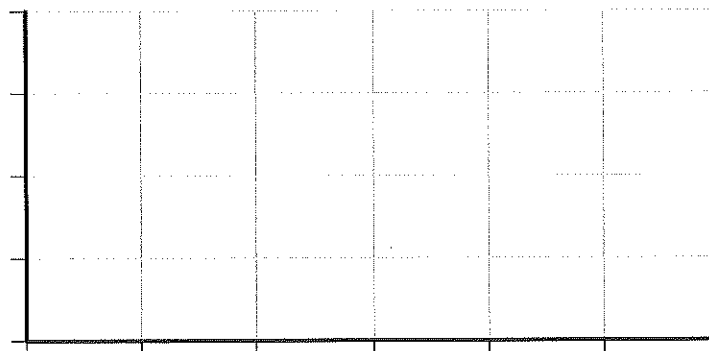
Mass water heated (g)	Applied voltage (V)	Current flowing (A)	Time current flowed (min)	Temperature rise (°C)	Input electrical power (W) $P_i = VI$	Input electrical energy (J) $E = Pt$	Heat energy absorbed by water $\Delta H = mc\Delta T$ (J)	Efficiency of coil (%) (See (a) below)
100	2	1.5	10	4.09				
100	4	1.5	10	7.66				
100	6	1.5	10	11.01				
100	8	1.5	10	13.86				
100	10	1.5	10	16.32				

- (a) Knowing that the efficiency of the coil can be found by comparing the heat energy absorbed by the water to the electrical energy supplied to it (expressed as a percentage), complete the missing values in the table and then calculate the efficiency of the coil for each applied voltage.
- (b) On the axes provided, draw a graph to show the relationship between the applied voltage and the input electrical energy to the coil.



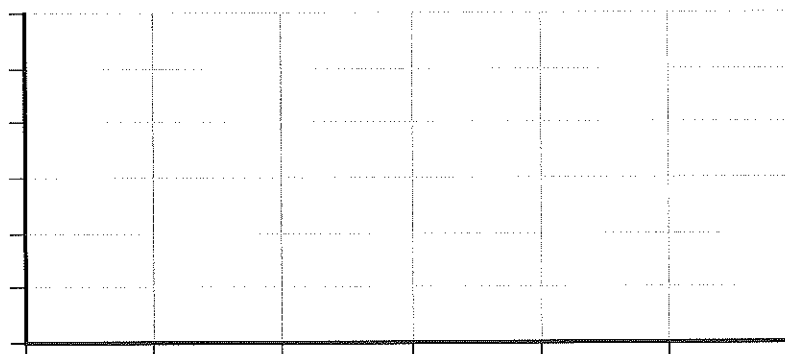
- (c) Write a conclusion from your graph.
-

- (d) On the axes provided, draw a graph to show the relationship between the input electrical energy and the heat energy produced by the coil.



- (e) Write a conclusion from your graph. Explain the nature of your conclusion.

- (f) On the axes provided, draw a graph to show the relationship between the input voltage and the efficiency of the coil.



- (g) Write a conclusion from your graph.

- (h) Outline the changes that would need to be made to the above experiment in order to determine the relationship between the energy produced by the heating coil and the current flowing through it.

3.4.3 Components in simple electric circuits.

3.4.3.1 Complete the table by drawing each of the electrical circuit components indicated.

Component	Diagram of component	Component	Diagram of component
Resistor		3.0 V battery	
Variable resistor		12 V battery	
Two resistors in series		Variable power supply	
Two resistors in parallel		DC power supply	
Fuse		AC power supply	
Ammeter		Coil	
Milliammeter		Transformer	
Microammeter		Closed switch	
Voltmeter		Open switch	
Galvanometer		Circuit wire	
Ohmmeter		Two circuit wires crossing but not connected	
Globe		Two circuit wires crossing and connected	
1.5 V cell		Earth	

3.4.4 Using ammeters.

3.4.4.1 Match the sentence halves to form complete sentences. Write your answers in the rows provided and choose from the alternate second halves given under the table.

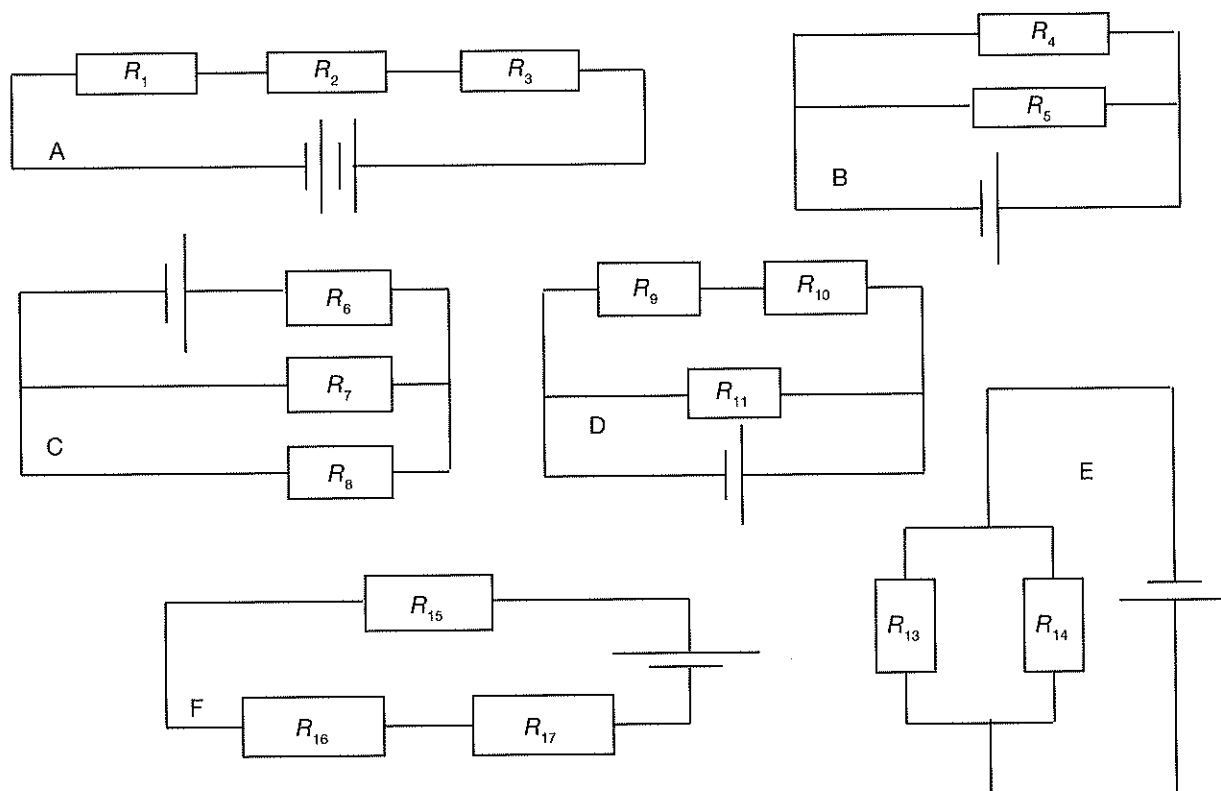
	Completed sentences
(a)	An electrical circuit is simply the device we use to change
(b)	So that we use the correct amount of electrical energy in a circuit, designers need to make
(c)	In this way, appliances connected into the circuit will not
(d)	It also ensures that the appliances have the current
(e)	At home some low voltage appliances
(f)	These are therefore connected via a transformer to
(g)	If they were connected directly to the power supply, the current flowing through
(h)	The instrument used to measure
(i)	Ammeters measure current flowing
(j)	Other current measuring instruments include
(k)	These are always connected into
(l)	In this way they measure the current flowing through that part of the circuit by
(m)	So that they don't affect the resistance of the circuit, it is essential

Second halves of sentences (not in correct order)

- (A) measurements so that the circuit is designed accurately.
- (B) them would be too large and they would burn out very quickly.
- (C) need only a small current.
- (D) electrical energy into other forms of energy.
- (E) circuits in series.
- (F) the power point in your home.

- (G) milliammeters and microammeters.
- (H) required for them to operate efficiently.
- (I) measuring the current flowing through themselves.
- (J) in amperes.
- (K) that ammeters have negligible resistance.
- (L) be damaged by excess current flow.
- (M) current is the ammeter.

- 3.4.4.2** Consider each of the following circuits. For each circuit, draw in ammeters that would enable you to measure the current flowing through the resistors. Using too many ammeters is just as incorrect as putting them in the wrong spot.



- 3.4.4.3** Complete the table.

Amps (A)	Milliamps (mA)	Microamps (μA)
0.1		
0.75		
2.6		
	200	
	1500	
	10 600	
		1000
		4×10^4
		2×10^6

3.4.5 Using voltmeters.

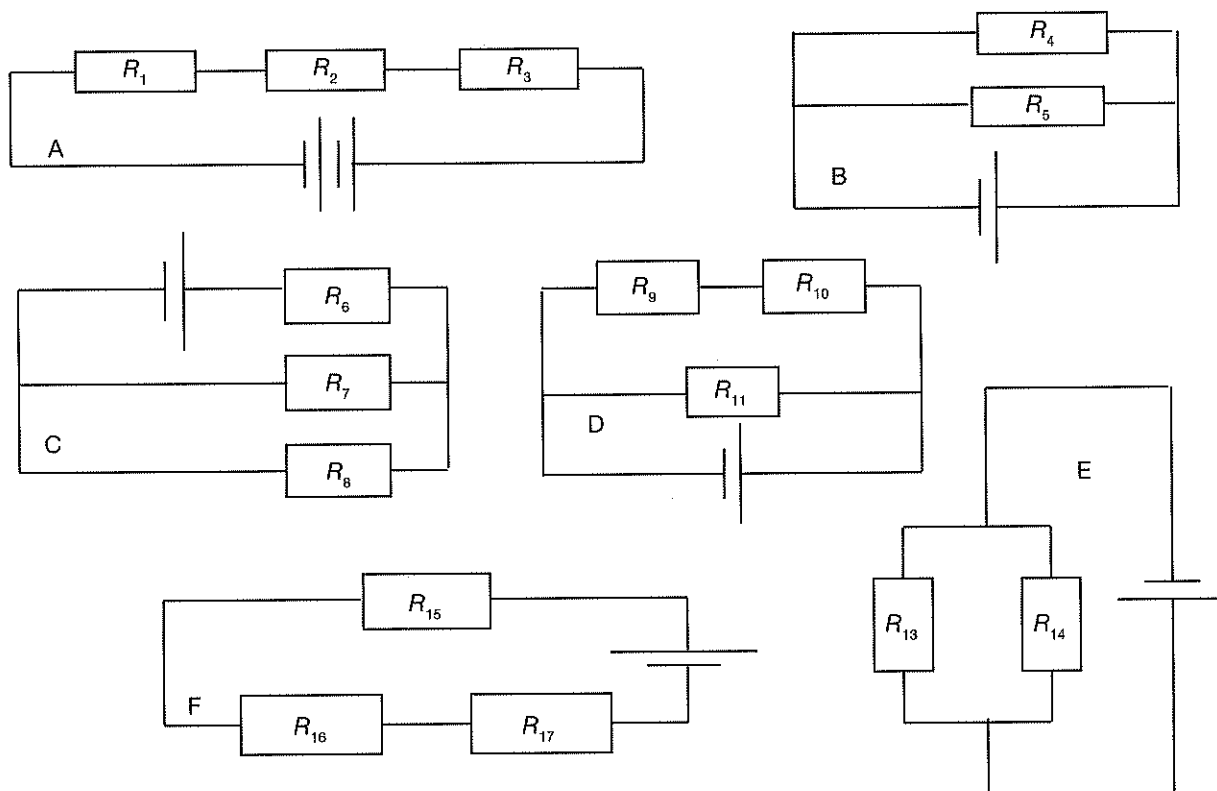
3.4.5.1 Complete each of the following sentences using your own words. The sequence of sentences should then be a summary of the work involved.

	Completed sentences
(a)	A battery is a source of
(b)	A battery is often described as an
(c)	A battery is a source of
(d)	Electrons flow through a circuit from a battery because of forces of
(e)	The ability of a battery to 'pump' electrons through a circuit is called
(f)	Voltage is a measure of how much energy a battery
(g)	The instrument used to
(h)	Voltmeters measure electrical potential difference across
(i)	To do this they must
(j)	Voltage can also be expressed as joules
(k)	Anything connected across two points in a circuit is said to be
(l)	The symbol for
(m)	Voltage is measured
(n)	The electrical potential difference between two points is equal to the work done
(o)	In order to minimise current through them, voltmeters have
(p)	This is required so that they do not affect the current flowing through

3.4.5.2

- (a) Recall how voltmeters are connected into a circuit.
- (b) Explain why they are connected into a circuit this way.
-
- (c) Predict the effect on the current flowing in a circuit if a voltmeter was connected in series with the power supply. Explain your answer.
-
-

3.4.5.3 Consider each of the following circuits. For each circuit, draw in voltmeters that would enable you to measure the potential difference across each resistor. Using too many voltmeters is just as incorrect as putting them in the wrong spot.



3.4.5.4 Describe each circuit so that your friends could draw it without referring to the circuit diagrams.

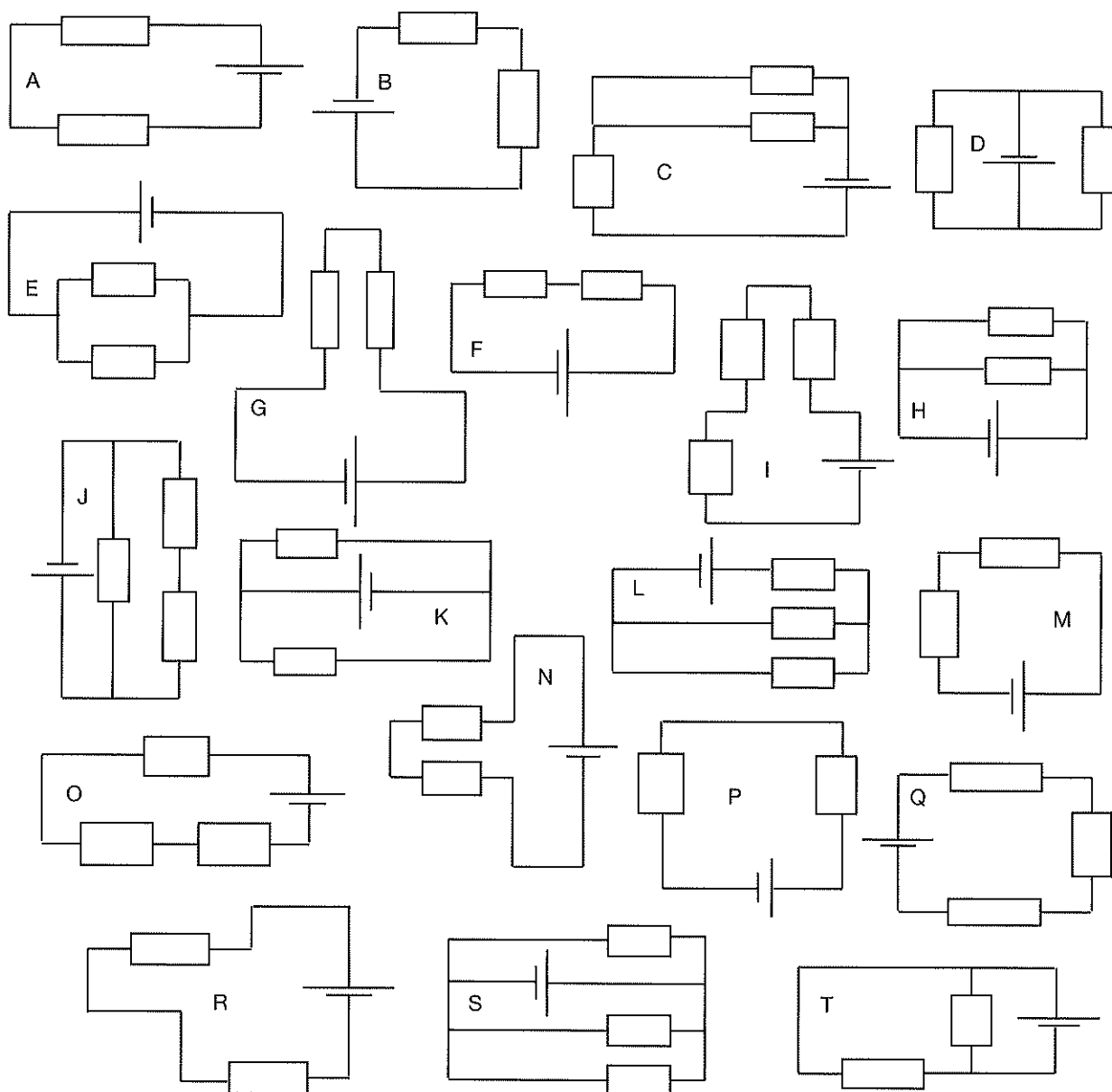
A	
B	
C	
D	
E	
F	

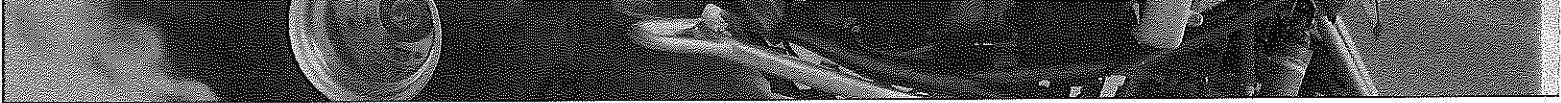
3.5 Circuit analysis and design.

3.5.1 Conductors in series and in parallel.

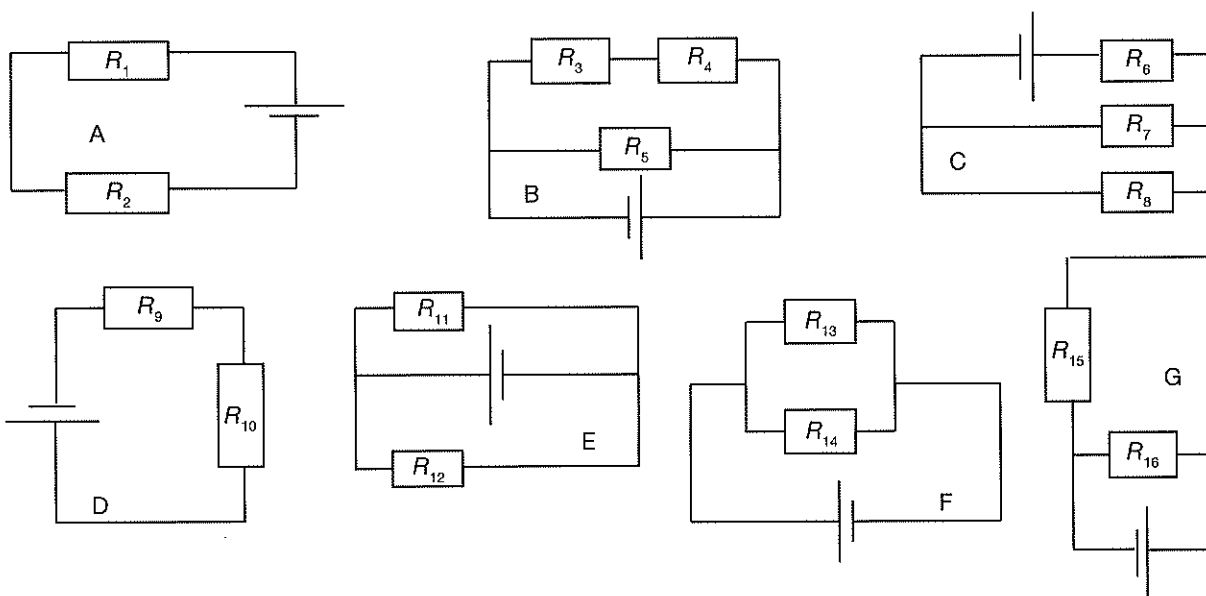
3.5.1.1 Classify each of the following circuits as having:

- Only resistors in series = S
- Only resistors in parallel = P
- Resistors in series and parallel = C





3.5.1.2 Describe each of the following circuits in such a way that your friend would be able to draw them without looking at anything except your description.

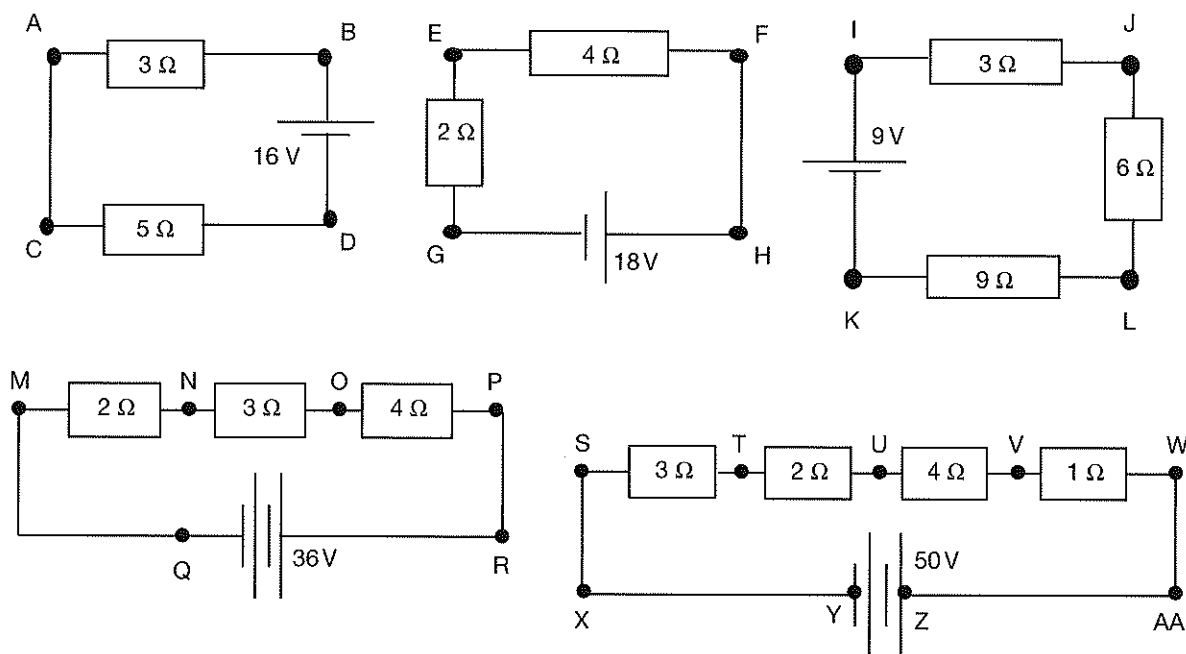


	Description of circuit
A	
B	
C	
D	
E	
F	
G	

3.5.2 Potential around a series circuit.

3.5.2.1

- (a) Determine the electrical potential at each labelled position in each of the circuits shown below. Write your answers in the table.
- (b) Use your answers to (a) to find the potential difference between the pairs of points indicated in the table.



Point	Potential	Point	Potential	Point	Potential	Point	Potential	Point	Potential
A		E		I		M		S	
B		F		J		N		T	
C		G		K		O		U	
D		H		L		P		V	
						Q		W	
						R		X	
BC		IL		YZ				Y	
FG		NQ		UX				Z	
GH		PQ		VS		SZ		AA	
JK		PM		WT		UY			

3.5.3 Analysing series circuits.

3.5.3.1

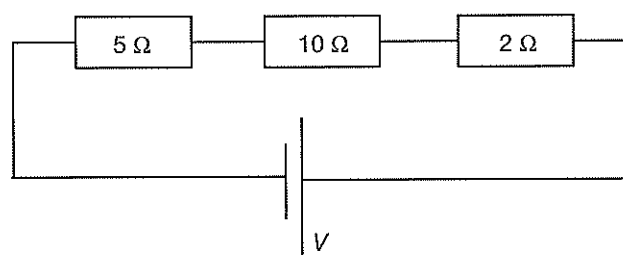
- (a) Draw a diagram to show a $3\ \Omega$, a $4\ \Omega$ and a $6\ \Omega$ resistor in series with each other.
- (b) What is the total resistance of this combination of resistors?

.....

.....

.....

- 3.5.3.2** A circuit contains three resistors, a $5\ \Omega$, a $10\ \Omega$ and a $2\ \Omega$ resistor, connected in series with each other as shown in the diagram. The power supply has a potential difference of V volts.



- (a) What is the total resistance of this circuit?

- (b) What is the ratio of the current flowing through the three resistors? $I_{5\ \Omega} : I_{10\ \Omega} : I_{2\ \Omega} =$

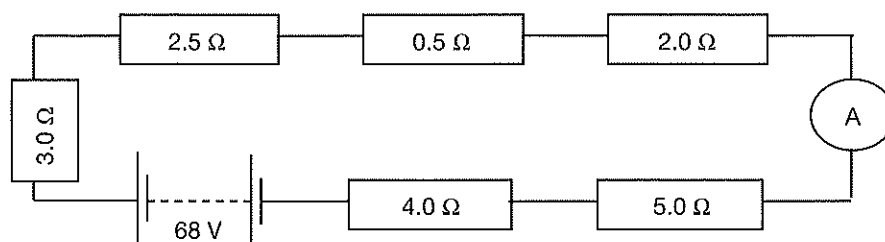
- (c) What is the ratio of the electrical potential difference across each resistor? $V_{5\ \Omega} : V_{10\ \Omega} : V_{2\ \Omega} =$

- 3.5.3.3** A $5\ \Omega$ conductor has a potential difference of $12.5\ \text{V}$ applied across it. Calculate the current flowing through it.

- 3.5.3.4** $24\ \text{V}$ applied across the ends of a conductor causes $1.6\ \text{A}$ to flow through it. Calculate the resistance of the conductor.

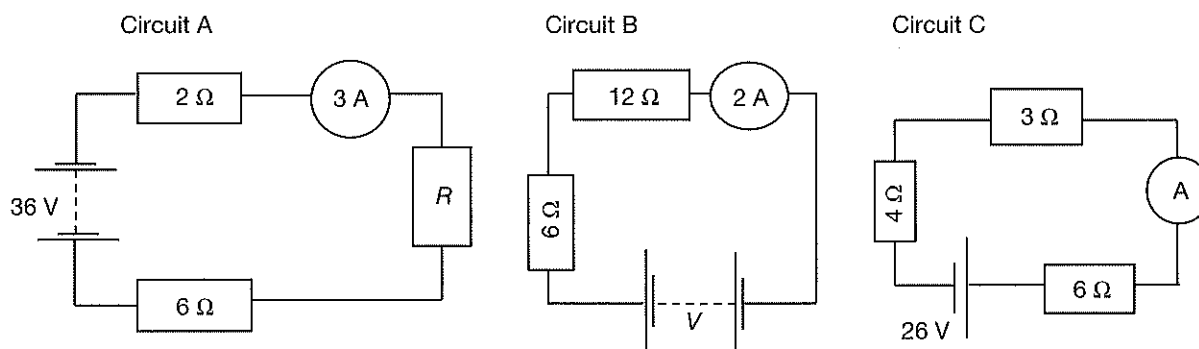
- 3.5.3.5** $3\ \text{A}$ flows through a length of conductor wire which has a resistance of $6\ \Omega$. Calculate the potential difference applied across the ends of the conductor.

3.5.3.6 Analyse the following circuit to complete the table.



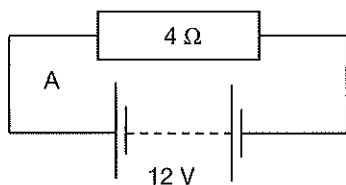
Resistor	0.5 Ω	2.0 Ω	2.5 Ω	3.0 Ω	4.0 Ω	5.0 Ω
Current through each resistor (A)						
Voltage across each resistor (V)						

3.5.3.7 Complete the table by calculating the values for the quantities indicated for each of the three circuits below.



	Circuit A			Circuit B		Circuit C		
Total circuit resistance (Ω)								
Circuit current (A)								
Current flowing through each resistor (A)	2 Ω	6 Ω	R Ω	6 Ω	12 Ω	3 Ω	4 Ω	6 Ω
Value of unknown resistor (Ω)								
Potential difference across each resistor (V)								
Potential of power source (V)								

3.5.3.8 For each of the circuits below, calculate values so that you can complete the answers.



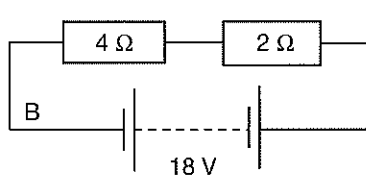
A

Total resistance =

Circuit current =

Current through $4\ \Omega$ =

Potential across $4\ \Omega$ =



B

Total resistance =

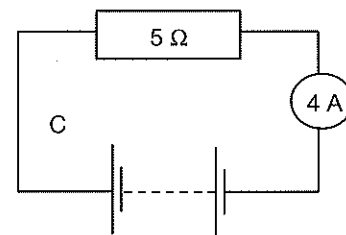
Circuit current =

Current through $2\ \Omega$ =

Current through $4\ \Omega$ =

Potential across $2\ \Omega$ =

Potential across $4\ \Omega$ =



C

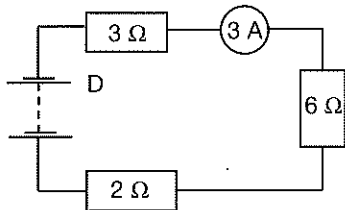
Total resistance =

Circuit current =

Current through $5\ \Omega$ =

Potential across $5\ \Omega$ =

Potential of source =



D

Total resistance =

Circuit current =

Current through $2\ \Omega$ =

Current through $3\ \Omega$ =

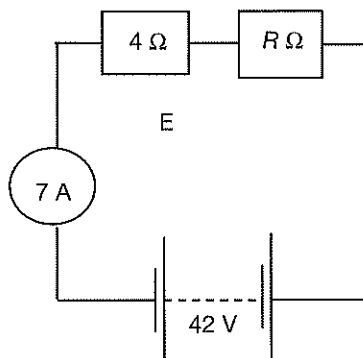
Current through $6\ \Omega$ =

Potential across $2\ \Omega$ =

Potential across $3\ \Omega$ =

Potential across $6\ \Omega$ =

Potential of source =



E

Total resistance =

Circuit current =

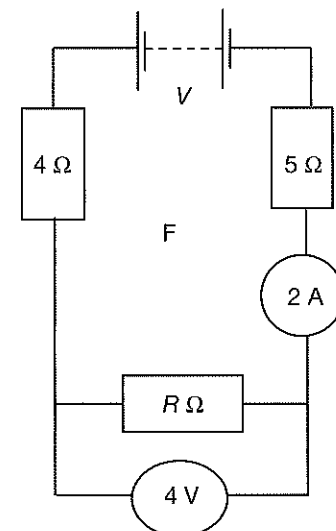
Current through $4\ \Omega$ =

Current through $R\ \Omega$ =

Potential across $4\ \Omega$ =

Potential across $R\ \Omega$ =

Value of resistor R =



F

Total resistance =

Circuit current =

Current through $4\ \Omega$ =

Current through $5\ \Omega$ =

Current through $R\ \Omega$ =

Potential across $4\ \Omega$ =

Potential across $5\ \Omega$ =

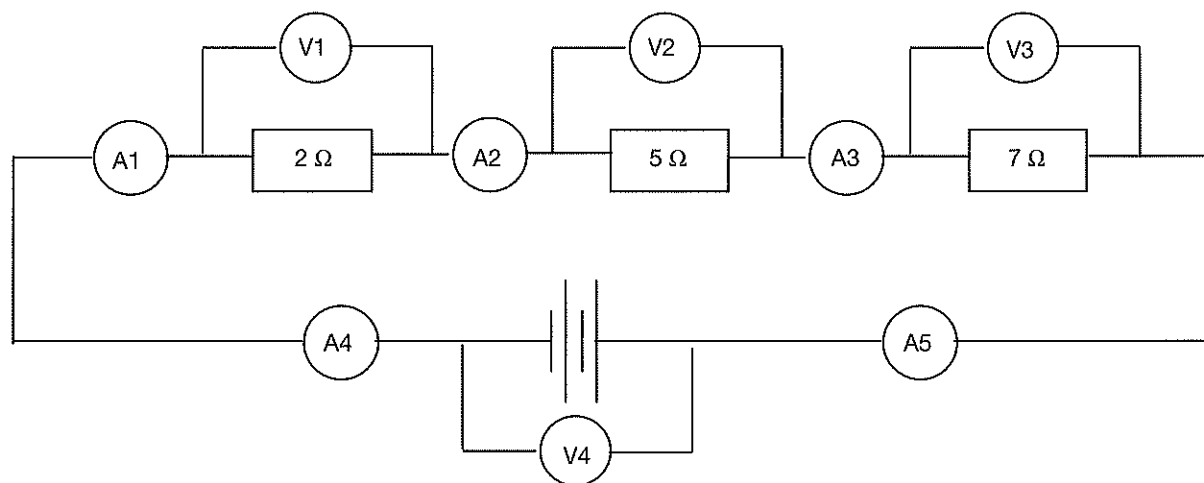
Potential across $R\ \Omega$ =

Potential of source =

Value of resistor R =

3.5.4 Series circuit – experimental analysis.

Students set up the circuit shown to study resistors in series. They connected voltmeters across the resistors and the power supply, and inserted ammeters in series at various positions in the circuit as shown. Their results are shown in the table. All voltages were measured in volts and all currents in amperes.



Results

Run	V1	V2	V3	V4	A1	A2	A3	A4	A5
1	3.4	8.8	12.3	25.0	1.7	1.8	1.9	1.8	1.8
2	3.4	8.6	12.2	24.0	1.8	1.8	1.7	1.7	1.8
3	3.6	8.8	12.4	24.4	1.7	2.7	1.7	1.8	1.8
4	3.5	8.7	12.5	24.5	1.7	1.8	1.6	1.6	1.6
Average									

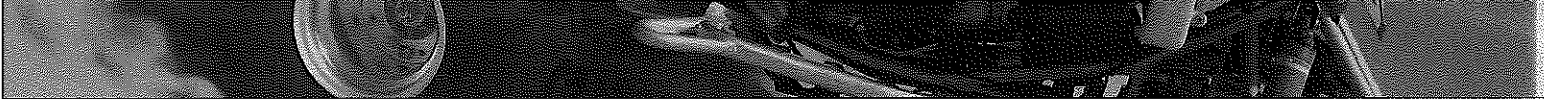
3.5.4.1 Complete the table by filling in the appropriate averages.

3.5.4.2 Comment on the process used to determine averages.

.....

.....

.....



3.5.4.3 Analyse these results to find the following.

- (a) The relationship between the current flowing through different resistors connected in series and the current flowing in the circuit.

.....

.....

- (b) The relationship between the potential differences across resistors connected in series and the potential difference of the power source.

.....

.....

- (c) The relationship between the total resistance of the circuit and the individual resistors.

.....

.....

- (d) The relationship between the potential difference across each resistor connected in series and the value of the resistor. Justify your answer.

.....

.....

- (e) The electrical potential at the position of each ammeter.

.....

.....

3.5.4.4 Complete the symbolic statements below to summarise the first three of these relationships (put + or – or = between each symbol).

For resistors connected in series:

I_{circuit} I_{R_1} I_{R_2} I_{R_3}

V_{circuit} V_{R_1} V_{R_2} V_{R_3}

R_{circuit} R_1 R_2 R_3

3.5.4.5 Justify one student's statement that the results of the experiment obey Ohm's law.

.....

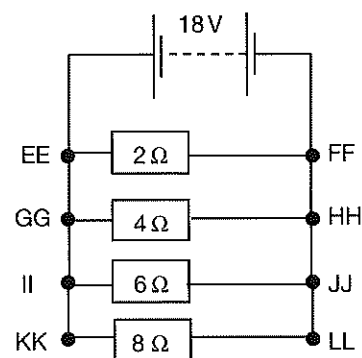
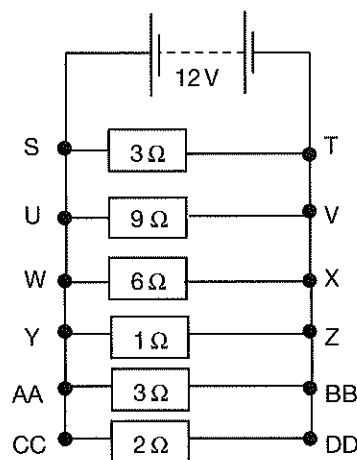
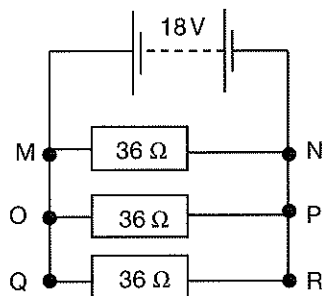
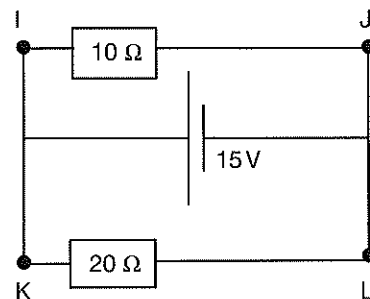
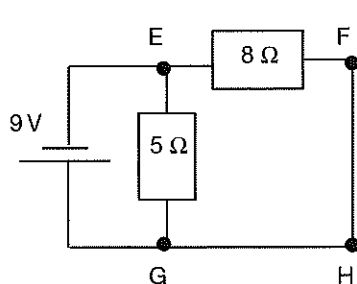
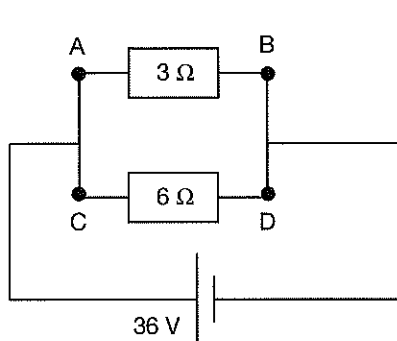
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3.5.5 Potential around a parallel circuit.

3.5.5.1 Determine the electrical potential at each labelled position in each of the circuits shown below. Write your answers in the table.



Point	Potential	Point	Potential	Point	Potential	Point	Potential	Point	Potential
A		I		Q		Y		FF	
B		J		R		Z		GG	
C		K		S		AA		HH	
D		L		T		BB		II	
E		M		U		CC		JJ	
F		N		V		DD		KK	
G		O		W		EE		LL	
H		P		X					

3.5.6 Analysing parallel circuits.

3.5.6.1

- (a) Draw a diagram to show a $3\ \Omega$, a $4\ \Omega$ and a $6\ \Omega$ resistor in parallel with each other.
- (b) What is the total resistance of this combination of resistors?

.....

.....

3.5.6.2 A circuit contains three $4\ \text{ohm}$ resistors, connected in parallel with each other as shown in the diagram. The power supply has a potential difference of V volts.

- (a) What is the ratio of the current flowing through the three resistors?

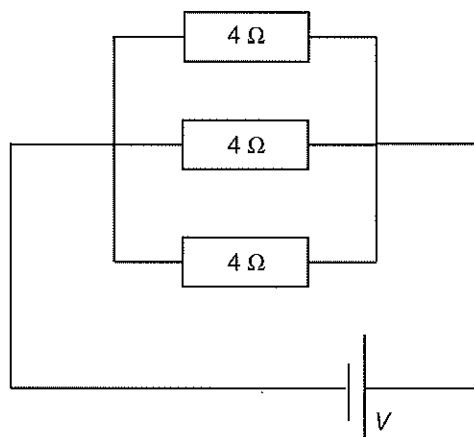
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.....

- (b) What is the ratio of the electrical potential difference across each resistor?

.....

.....



3.5.6.3 A circuit contains three resistors, a $3\ \text{ohm}$, a $6\ \text{ohm}$ and a $9\ \text{ohm}$ resistor, connected in parallel with each other as shown in the diagram. The power supply has a potential difference of V volts.

- (a) What is the ratio of the current flowing through the three resistors?

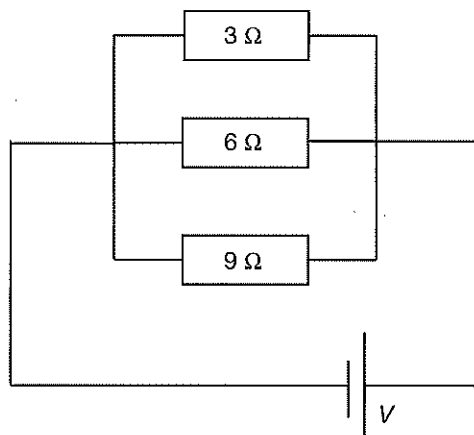
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.....

- (b) What is the ratio of the electrical potential difference across each resistor?

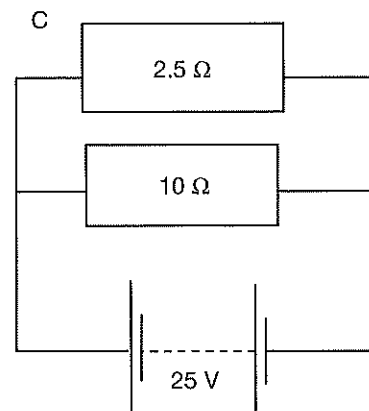
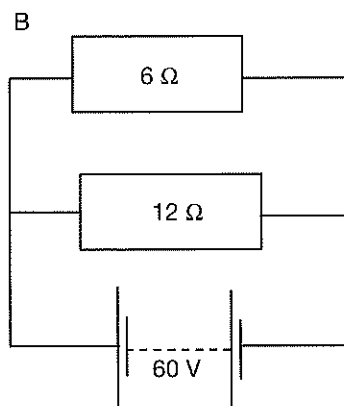
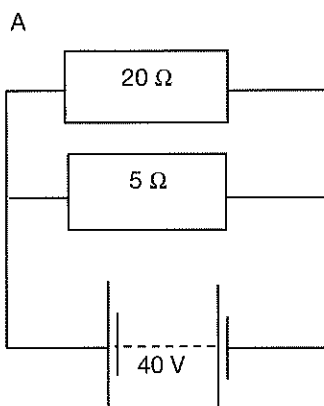
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3.5.6.4 In the space provided draw a diagram of a circuit which has two resistors in parallel and a total resistance of $4\ \text{ohms}$. The circuit contains a power source and $6\ \text{A}$ of current flows in the circuit.

3.5.6.5 Calculate the indicated quantities for each of the following circuits.



A

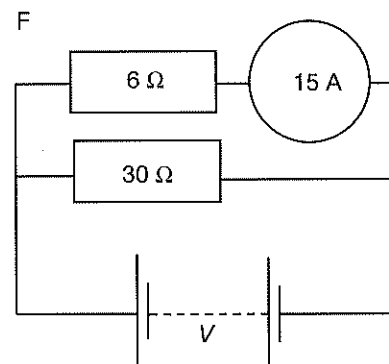
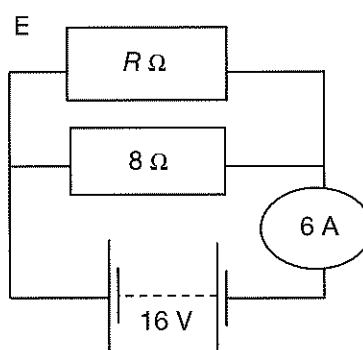
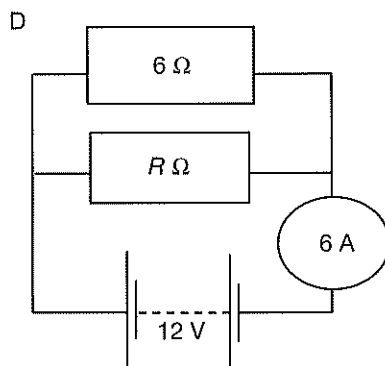
Total resistance =
 Circuit current =
 Current through $5\ \Omega$ =
 Current through $20\ \Omega$ =
 Potential across $5\ \Omega$ =
 Potential across $20\ \Omega$ =

B

Total resistance =
 Circuit current =
 Current through $6\ \Omega$ =
 Current through $12\ \Omega$ =
 Potential across $6\ \Omega$ =
 Potential across $12\ \Omega$ =

C

Total resistance =
 Circuit current =
 Current through $2.5\ \Omega$ =
 Current through $10\ \Omega$ =
 Potential across $2.5\ \Omega$ =
 Potential across $10\ \Omega$ =



D

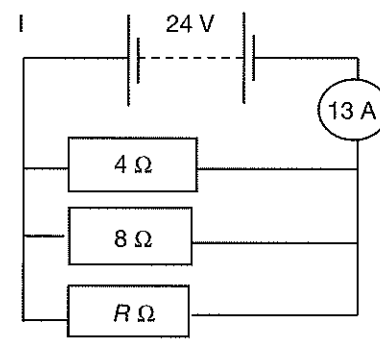
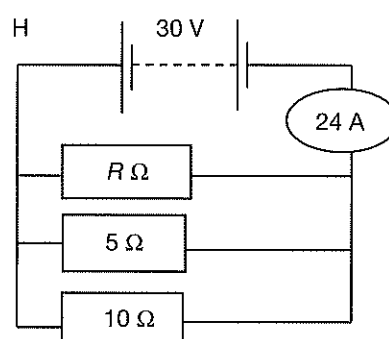
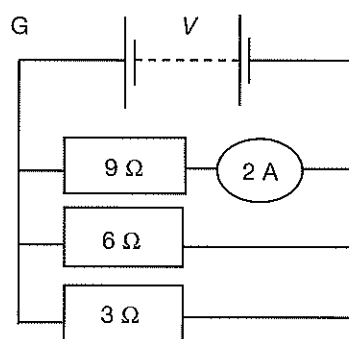
Total resistance =
 Circuit current =
 Current through $6\ \Omega$ =
 Current through $R\ \Omega$ =
 Potential across $6\ \Omega$ =
 Potential across $R\ \Omega$ =
 Value of resistor R =

E

Total resistance =
 Circuit current =
 Current through $R\ \Omega$ =
 Current through $8\ \Omega$ =
 Potential across $R\ \Omega$ =
 Potential across $8\ \Omega$ =
 Value of resistor R =

F

Total resistance =
 Circuit current =
 Current through $6\ \Omega$ =
 Current through $30\ \Omega$ =
 Potential across $6\ \Omega$ =
 Potential across $30\ \Omega$ =



G

Total resistance =
 Circuit current =
 Current through $9\ \Omega$ =
 Current through $6\ \Omega$ =
 Current through $3\ \Omega$ =
 Potential across $9\ \Omega$ =
 Potential across $6\ \Omega$ =
 Potential across $3\ \Omega$ =
 Potential of source =

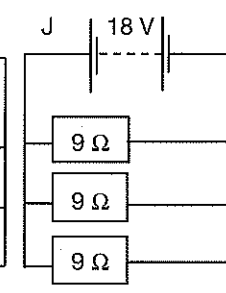
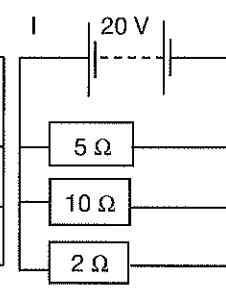
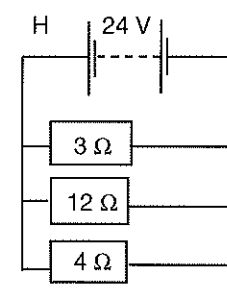
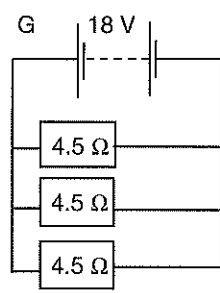
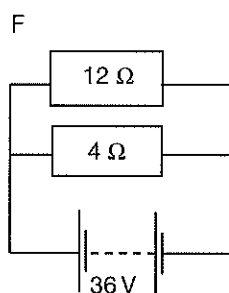
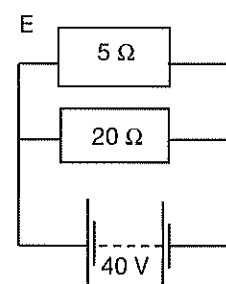
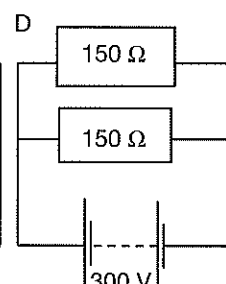
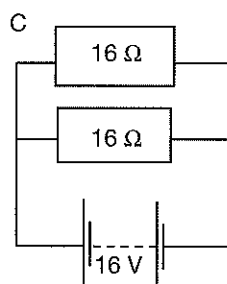
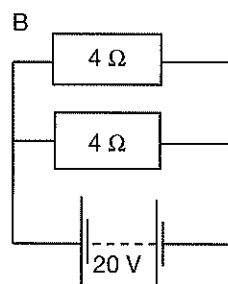
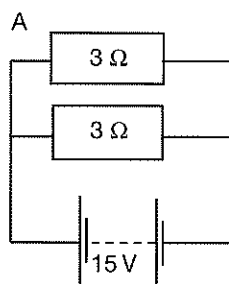
H

Total resistance =
 Circuit current =
 Current through $R\ \Omega$ =
 Current through $5\ \Omega$ =
 Current through $10\ \Omega$ =
 Potential across $R\ \Omega$ =
 Potential across $5\ \Omega$ =
 Potential across $10\ \Omega$ =
 Value of resistor R =

I

Total resistance =
 Circuit current =
 Current through $4\ \Omega$ =
 Current through $8\ \Omega$ =
 Current through $R\ \Omega$ =
 Potential across $4\ \Omega$ =
 Potential across $8\ \Omega$ =
 Potential across $R\ \Omega$ =
 Value of resistor R =

3.5.6.6 Refer to the following circuits to answer this question.



- (a) What is the total resistance of each circuit?

Circuit	A	B	C	D	E	F	G	H	I	J
Total resistance (Ω)										

- (b) Compare your answer for circuits A, B, C and D and suggest another way of determining the total resistance of a circuit which contains two identical resistors in parallel.

.....

.....

- (c) Compare your answer for circuits G and J and suggest another way of determining the total resistance of a circuit which contains three identical resistors in parallel.

.....

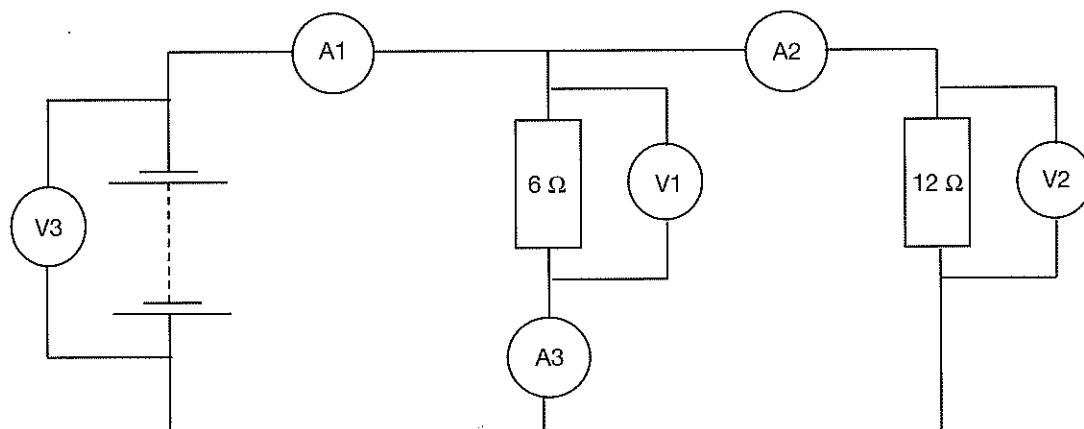
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- (d) Given that the potential difference across resistors connected in parallel is the same, and since there are no series resistors in any of the circuits, this potential difference equals the potential difference of the power source. Predict the current flowing through each resistor.

		Resistor	A	B	C	D	E	F	G	H	I	J
Current in resistor (A)		2 Ω										
		3 Ω										
		4 Ω										
		4.5 Ω										
		5 Ω										
		9 Ω										
		10 Ω										
		12 Ω										
		16 Ω										
		20 Ω										
		150 Ω										

3.5.7 Parallel circuit – experimental analysis.

Students set up the circuit shown to study resistors in parallel. In separate runs they connected voltmeters and ammeters as shown in the diagram and recorded the reading on each. Their results are shown in the table. All voltages were measured in volts and all currents in amperes.



Run	V1	V2	V3	A1	A2	A3
1	12.3	11.6	12.4	2.9	1.0	1.9
2	11.7	12.3	11.8	2.9	0.8	1.8
3	12.0	12.1	11.7	3.3	1.1	2.2
4	11.9	11.9	11.9	3.1	1.0	2.0
5	12.1	12.1	12.1	2.8	1.1	2.1
Average						

3.5.7.1 Complete the table.

3.5.7.2 Analyse these results to find the following.

(a) The relationship between the current flowing through different resistors connected in parallel and the current flowing in the circuit.

(b) The relationship between the potential differences across resistors connected in parallel and the potential difference of the power source.

(c) The total resistance of the circuit.

(d) The relationship between the potential differences across resistors connected in parallel and the value of the resistor.

(e) The electrical potential at the position of each ammeter.

- 3.5.7.3** Complete the symbolic statements to summarise the first three of these relationships (put + or – or = between each symbol).

For resistors connected in parallel:

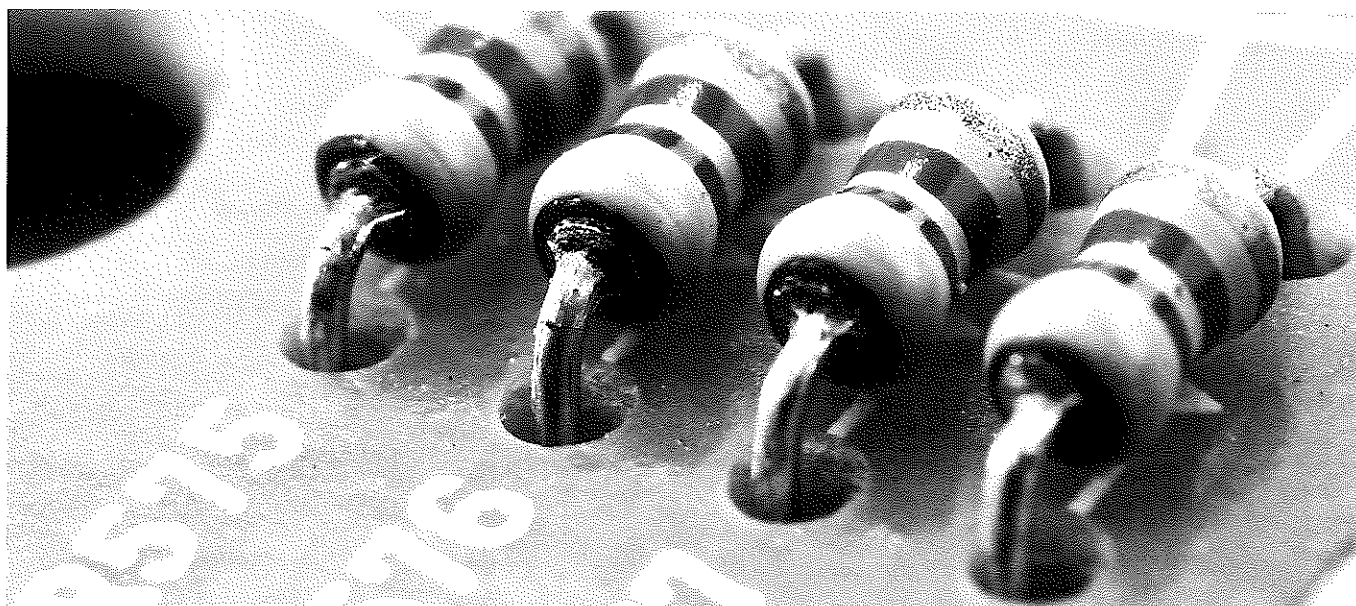
I_{circuit} I_{R_1} I_{R_2} I_{R_3}

V_{circuit} V_{R_1} V_{R_2} V_{R_3}

- 3.5.7.4** Show, that at least for this circuit, the total resistance (R_T) of a combination of two resistors in parallel is given by the following.

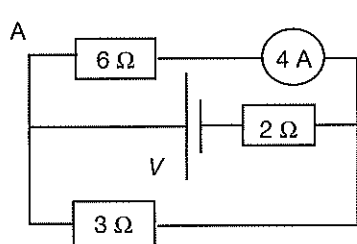
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \quad \text{or} \quad R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

- 3.5.7.5** In what way did the students doing this experiment attempt to determine the reliability of their results?



3.5.8 Analysing circuits with series and parallel resistors.

3.5.8.1 Analyse each of the following circuits to determine values for the quantities indicated.



A

Total resistance =

Circuit current =

Current through $6\ \Omega$ =

Current through $2\ \Omega$ =

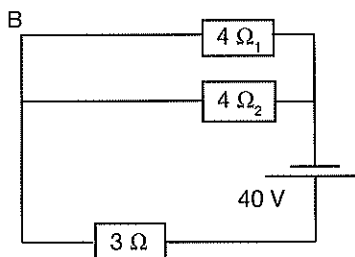
Current through $3\ \Omega$ =

Potential across $6\ \Omega$ =

Potential across $2\ \Omega$ =

Potential across $3\ \Omega$ =

Potential of source =



B

Total resistance =

Circuit current =

Current through $3\ \Omega$ =

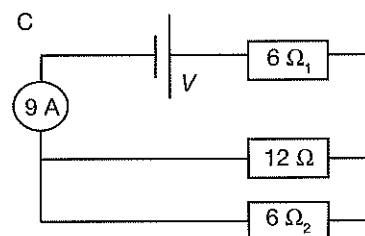
Current through $4\ \Omega_1$ =

Current through $4\ \Omega_2$ =

Potential across $3\ \Omega$ =

Potential across $4\ \Omega_1$ =

Potential across $4\ \Omega_2$ =



C

Total resistance =

Circuit current =

Current through $6\ \Omega_1$ =

Current through $12\ \Omega$ =

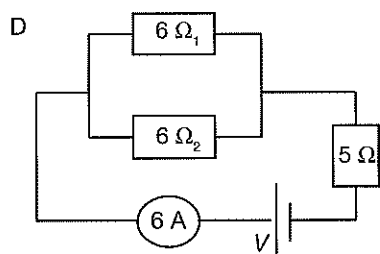
Current through $6\ \Omega_2$ =

Potential across $6\ \Omega_1$ =

Potential across $12\ \Omega$ =

Potential across $6\ \Omega_2$ =

Potential of source =



D

Total resistance =

Circuit current =

Current through $6\ \Omega_1$ =

Current through $6\ \Omega_2$ =

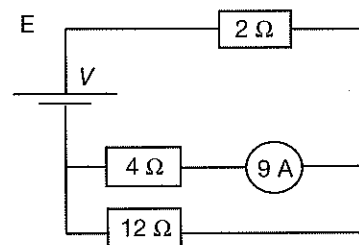
Current through $5\ \Omega$ =

Potential across $6\ \Omega_1$ =

Potential across $6\ \Omega_2$ =

Potential across $5\ \Omega$ =

Potential of source =



E

Total resistance =

Circuit current =

Current through $2\ \Omega$ =

Current through $4\ \Omega$ =

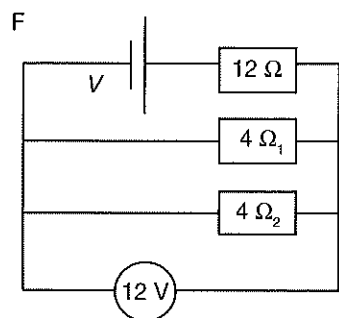
Current through $12\ \Omega$ =

Potential across $2\ \Omega$ =

Potential across $4\ \Omega$ =

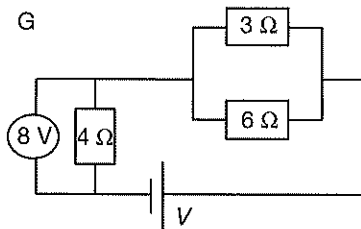
Potential across $12\ \Omega$ =

Potential of source =



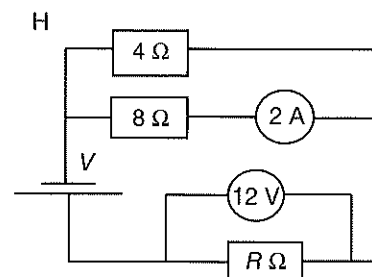
F

Total resistance =
 Circuit current =
 Current through $12\ \Omega$ =
 Current through $4\ \Omega_1$ =
 Current through $4\ \Omega_2$ =
 Potential across $12\ \Omega$ =
 Potential across $4\ \Omega_1$ =
 Potential across $4\ \Omega_2$ =
 Potential of source =



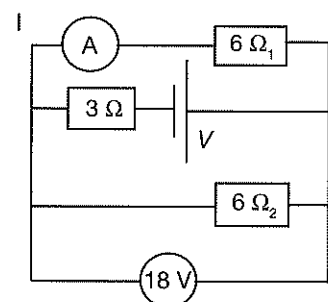
G

Total resistance =
 Circuit current =
 Current through $3\ \Omega$ =
 Current through $6\ \Omega$ =
 Current through $4\ \Omega$ =
 Potential across $3\ \Omega$ =
 Potential across $6\ \Omega$ =
 Potential across $4\ \Omega$ =
 Potential of source =



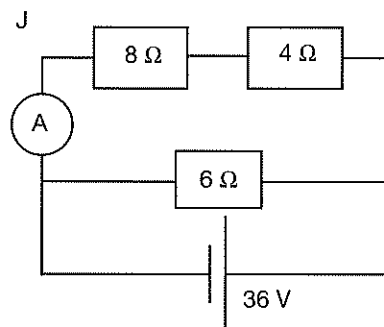
H

Total resistance =
 Circuit current =
 Current through $4\ \Omega$ =
 Current through $8\ \Omega$ =
 Current through $R\ \Omega$ =
 Potential across $4\ \Omega$ =
 Potential across $8\ \Omega$ =
 Potential across $R\ \Omega$ =
 Potential of source =
 Value of resistor R =



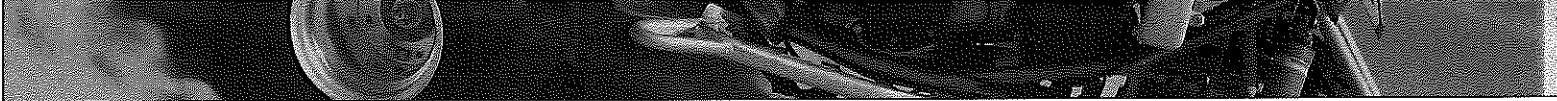
I

Total resistance =
 Circuit current =
 Current through $6\ \Omega_1$ =
 Current through $6\ \Omega_2$ =
 Current through $3\ \Omega$ =
 Potential across $6\ \Omega_1$ =
 Potential across $6\ \Omega_2$ =
 Potential across $3\ \Omega$ =
 Potential of source =
 Reading on meter A =



J

Total resistance =
 Circuit current =
 Current through $8\ \Omega$ =
 Current through $4\ \Omega$ =
 Current through $6\ \Omega$ =
 Potential across $8\ \Omega$ =
 Potential across $4\ \Omega$ =
 Potential across $6\ \Omega$ =
 Potential of source =
 Reading on meter A =



3.6 Transferring and transforming electrical energy.

3.6.1 Transformers.

Note: Transformers are not mentioned in the syllabus until Unit 3. These questions are included for completeness of the concept only and for schools that introduce them in their teaching program at this stage.

3.6.1.1

- (a) Explain the function of transformers.
.....
- (b) Name the two different types of transformers, and state the specific function of each type.
.....
.....
- (c) In terms of their physical structures, explain how the two types of transformers are designed to do their job.
.....
.....
- (d) State the relationship between the voltage input and output in a transformer to the numbers of turns in the input and output coils of the transformer.
.....
.....

3.6.1.2 A transformer will not work with DC current. Explain why.
.....
.....

3.6.1.3 A transformer has a 1000 turn primary coil. It is to be used to convert 240 V AC to 120 V AC.

- (a) Identify this as a step-up or step-down transformer.
- (b) What is the number of coils in the secondary coil of this transformer?

3.6.1.4 The output coil of a transformer has five times as many coils as the input coil.

- (a) Identify this as a step-up or step-down transformer.
- (b) If the input voltage was 100 V, state the output voltage.
- (c) If the input current was 15 A, state the output current.

3.6.1.5 A 240 V battery charger is used to charge 1.5 V batteries through a 12 V transformer. It has 3600 turns in the input coil. Assume 100% efficiency.

- (a) What is the number of turns in the output coil?
- (b) If the input current is 5 mA, calculate the output current.

3.6.1.6 A transformer changes 240 V to 30 000 V.

- (a) Calculate the ratio of the number of turns in each coil.
- (b) Calculate the ratio of the currents in each coil.
- (c) Identify the output coil.

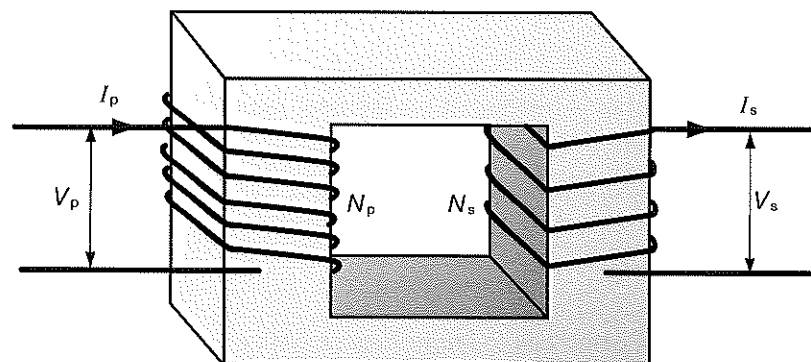
3.6.1.7 A commercial transformer (which has other circuit components to do other electrical 'jobs' built into it) has these characteristics printed on it:

Input AC 240 V, 50 Hz, 10 A

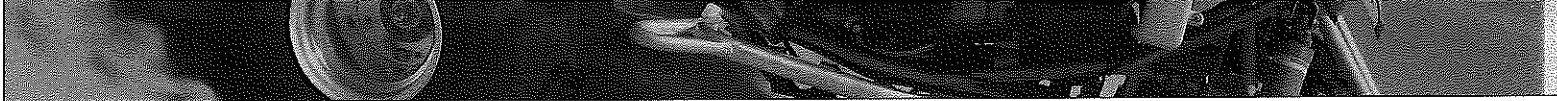
Output DC 12 V, 0.5 A

- (a) Explain each of these characteristics.
.....
.....
- (b) Identify this as a step-up or step-down transformer.
- (c) State what else this transformer's other components are doing.
- (d) If the input coil has 2000 turns, deduce the number in the output coil.
- (e) Calculate what the output current should be, based on the voltage figures.
- (f) Explain why the output current is rated at only 0.5 A.
.....

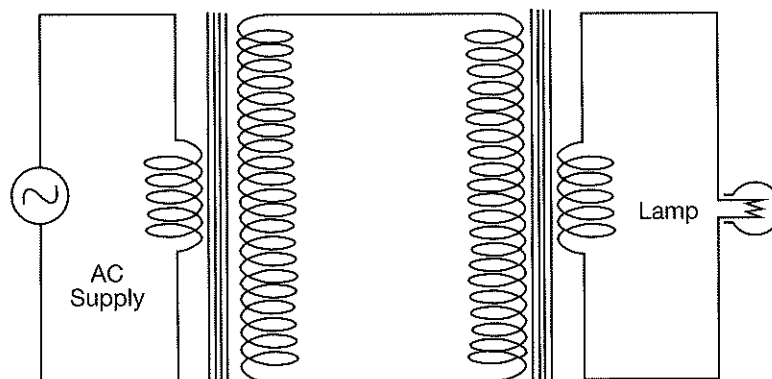
3.6.1.8 The diagram shows a transformer.



- (a) Identify this as a step-up or step-down transformer.
- (b) Justify your answer.
.....
- (c) If this transformer has 360 turns and 18 turns in its coils, identify each coil.
.....
- (d) If the input voltage was 220 V, calculate the output voltage.



3.6.1.9 The diagram shows two transformers connected into a circuit.



- (a) Assuming the coils are drawn to scale, compare the input AC current with the current in the lamp.
.....
- (b) Compare the input AC voltage with the voltage across the lamp.
.....
- (c) State whether the power loss in transmission lines depends on the voltage across them or the current in them.
.....
- (d) Assess the value of having a transformer like this connected into a circuit.
.....
- (e) What type of transformer is needed at a substation which distributes electrical power to suburban homes? Justify your answer.
.....
.....
.....

3.6.1.10 Without transformers, electricity in our homes would be costly. It would certainly be very differently supplied. Explain why.

.....
.....
.....
.....
.....
.....

3.6.2 Transforming electrical energy.

3.6.2.1 Complete the table by listing four devices that transform electrical energy into other useful forms of energy for use in your kitchen at home.

Alongside each, write the useful form of energy that device produces. Try to choose four devices that produce four *different* types of useful energy.

Device	Useful form of energy produced

3.6.2.2 The production of heat energy in conductors can sometimes be a desired outcome and sometimes an undesired outcome.

Complete the table to indicate whether the devices listed produce heat energy as a useful or a non-useful product, and to state the most useful form(s) of energy produced by each device.

Device	Heat energy useful or non-useful?	Most useful form(s) of energy produced
Dishwasher		
TV		
Refrigerator motor		
Frypan		
Microwave oven		
Motor in clothes dryer		

3.6.2.3 The energy transformations that occur in a typical radio and TV broadcasting station are summarised in the passage below.

In both radio and television broadcasting, microphones change sound energy into electrical energy. This electrical energy is changed by the transmitters into electromagnetic radiation. These are reconverted into electrical energy by receiving antennas and then into kinetic energy in vibrating speaker cones which produce the soundwaves we hear.

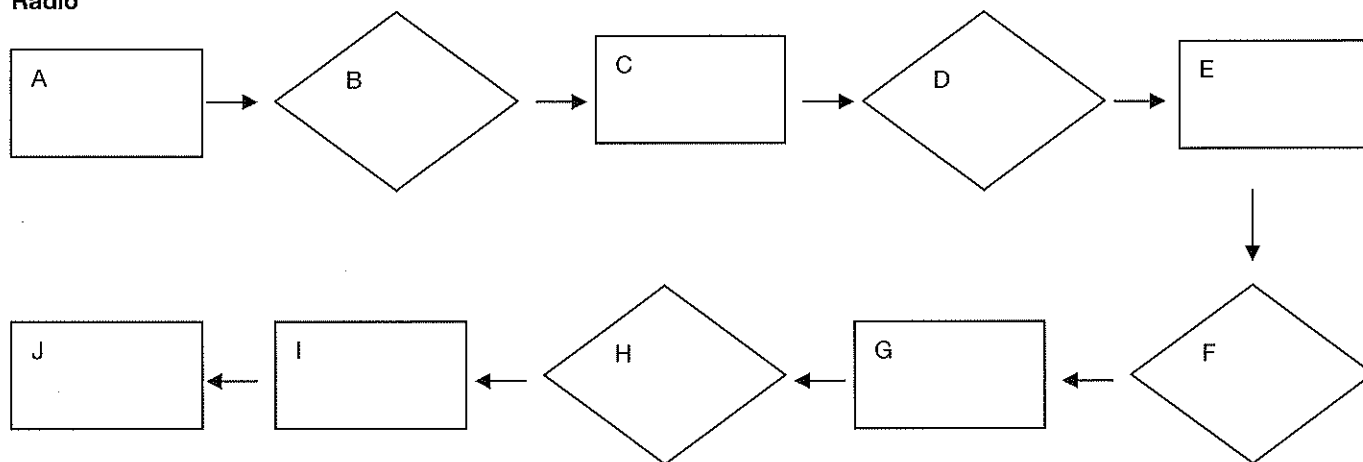
The picture information for televisions requires a separate converter (a video camera) to change light energy into electrical impulses that are then changed into electromagnetic signals and transmitted with the signals carrying the sound. Again the antenna reconverts these to electrical energy that is sent to the TV set. The television changes the electrical energy into kinetic energy carried by electrons from the electron gun at the back of the set to the screen. When the electrons hit phosphorescent chemicals on the screen, their kinetic energy is converted back into light energy.

Complete the diagrams below to summarise the *energy* changes that occur in radio and television transmission and reception.

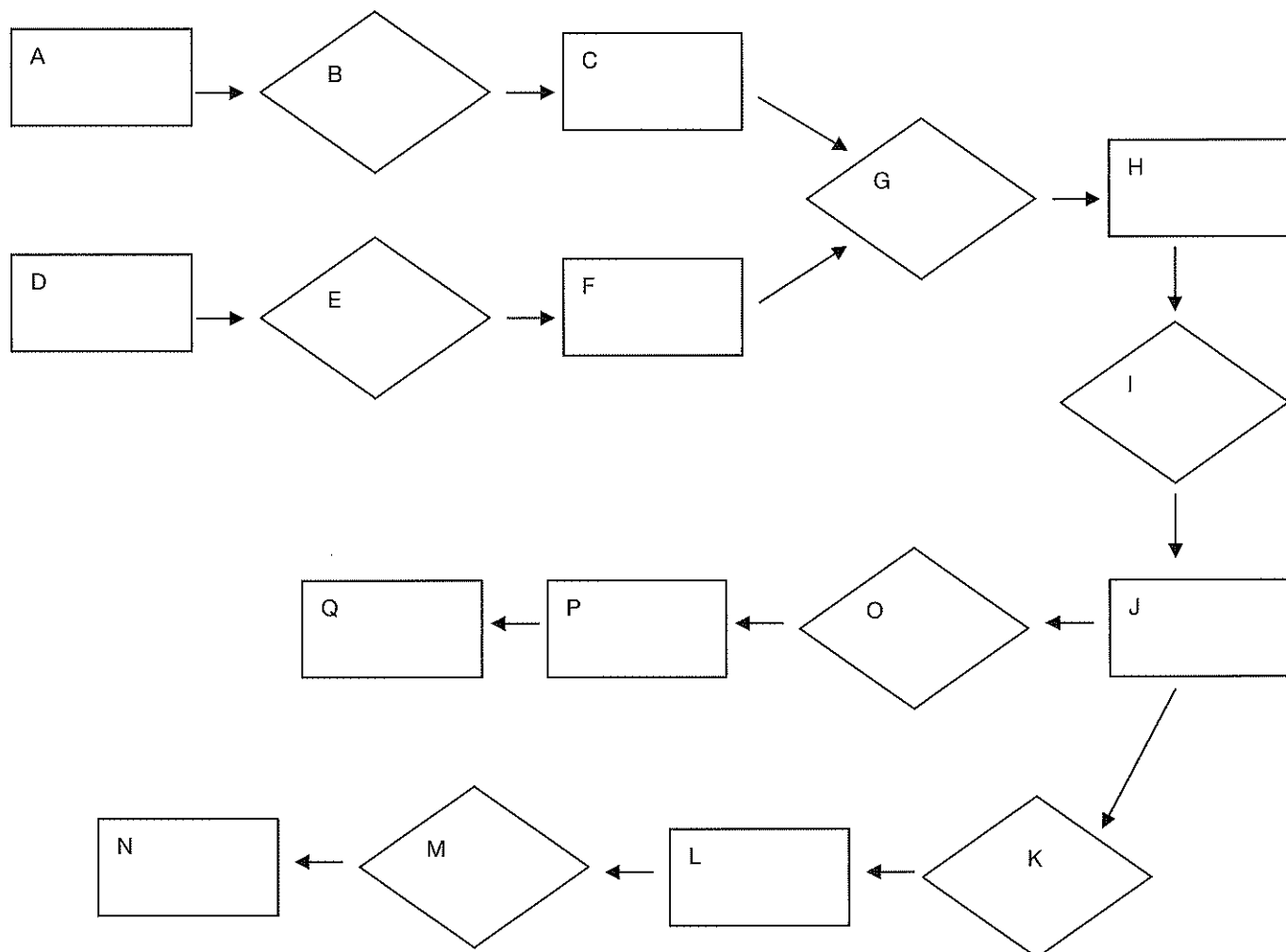
The diamond shapes are included for the names of the converting devices.

Write the forms of energy in the rectangles.

Radio



Television



3.6.3 Other problems with transmitting.

3.6.3.1

- (a) Explain why it is necessary to transmit alternating current electricity from power stations to wherever it is used.

- (b) Why can't direct current electricity be transmitted in the same way?

- (c) What is the major source of energy loss when transmitting electricity?

3.6.3.2

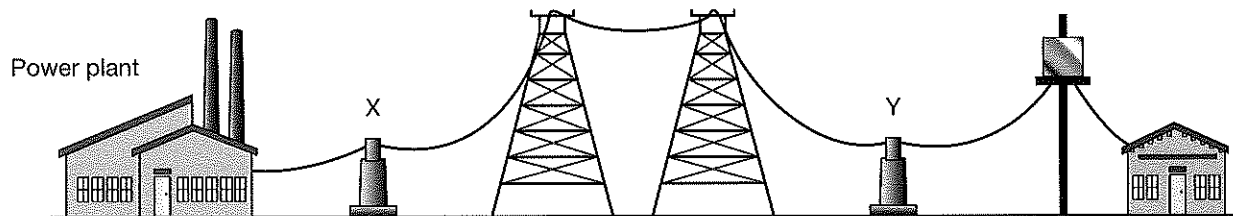
- (a) Complete the table to calculate energy losses during electricity transmission. Assume the power station is producing 40 MW of power, and needs to transmit it 80 km through 3 ohms km⁻¹ lines.

Transmission voltage (V)	100 000	250 000	350 000	500 000
Power generated (MW)	40	40	40	40
Current in lines $I = \frac{P}{V}$ (A)				
Total resistance of lines (Ω)				
Power loss in lines $P = I^2 R$ (MW)				
Power left after losses (MW)				
Power lost (%)				
Power getting to consumers (%)				

- (b) What are the implications for you and your family, of the transmission of electricity using low line voltages? Justify your answer.

- (c) Larger energy savings could be made by transmitting at even higher than 500 000 V, but this is usually the limit. Explain why higher transmission voltages are not used.

- 3.6.3.3** The diagram shows the position of two transformers in a transmission line from the power station to homes.



- (a) Which of the two transformers is most likely a step-up transformer? Justify your answer.
-
-
- (b) Which of the two transformers is most likely a step-down transformer? Justify your answer.
-

- 3.6.3.4** Complete the table to summarise safety/protection features used in high voltage transmission systems and the brief descriptions of their functions.

Structure	How this protects against lightning strikes
Shield conductors	
	Saucer-shaped stacks of ceramic material used to insulate transmission lines from the tower and each other at lengths depending on the voltage to prevent electrical discharges between wires and the tower.
	Runs from the top of the pole down into the earth to carry lightning strikes to earth thus protecting the tower and the transmission lines.
Metal tower	
Distance between towers	

3.6.4 Circuits in homes.

3.6.4.1 Why are transformers needed in our homes?

- (A) To increase the current supply to appliances.
- (B) To reduce voltages to safe levels.
- (C) To cater for different voltage input appliances.
- (D) To reduce the amount of electrical power the appliance uses.

3.6.4.2 Which is an advantage of an electrical appliance that works on low voltages?

- (A) They are safer to use.
- (B) They consume more power.
- (C) They work on a high current.
- (D) They can be used outside in the weather.

3.6.4.3 Many electrical toys used by children are low voltage toys. Which choice best explains this?

- (A) They cost less to build and buy.
- (B) They are more reliable than high voltage toys.
- (C) They cost less to run.
- (D) They are safer if some fault develops.

3.6.4.4 Most appliances used by adults are 240 V rather than low voltage appliances. Which choice best explains this?

- (A) Adults are more careful with electricity than children.
- (B) High voltage appliances are more reliable than low voltage appliances.
- (C) High voltage appliances develop more power and do the work more quickly.
- (D) High voltage appliances cost less to run than low voltage appliances.

3.6.4.5 A lighting circuit in a home has a 10 A fuse connected into it. What is the maximum number of 80 W light bulbs that could be connected and all on at the same time?

- (A) 28 (B) 29 (C) 30 (D) 31

3.6.4.6 What is the total wattage of appliances that can be used in a 15 A circuit simultaneously?

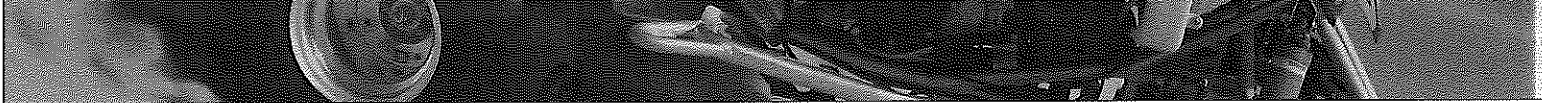
- (A) 16 W
- (B) 240 W
- (C) 2400 W
- (D) 3600 W

3.6.4.7 Older homes have fuses rather than circuit breakers. Which statement about fuses and circuit breakers is correct?

- (A) Circuit breakers are safer because they work faster.
- (B) Fuses are safer because they work faster.
- (C) Circuit breakers work by a wire overheating, melting and then breaking the circuit.
- (D) Fuses are reset by simply pressing a reset button in the fuse box.

3.6.4.8 If one globe in a string of 200 fairy lights blows, then 20 bulbs go out. Why?

- (A) The 200 globes are all connected in series with each other.
- (B) The 20 lights that go out are in parallel with the globe that blows.
- (C) The string of 200 lights is made up of 10 series strings of 20 globes all connected in parallel with each other.
- (D) When the globe blows, a current surge can cause the others to blow also.



3.6.4.9

The power circuit in a home is designed to carry a maximum of 15 A. In the kitchen there are several appliances as listed in the table.

Appliance	Voltage of appliance (V)	Power appliance develops (W)	Current drawn by appliance (A)
Coffee machine	240	1200	
Frypan	240	1800	
Kettle	240	1500	
Laptop	18		4.2
Microwave	240		2.5
Rice cooker	240		2.083
Toaster	240		3.33

- (a) Complete the table.
- (b) What is the maximum number of these appliances that could be used at the same time? Justify your answer.

.....

.....

.....

.....

- (c) What is the minimum number of appliances that could be safely used at the same time? Justify your answer.

.....

.....

3.6.4.10

- (a) Where are parallel circuits used in homes? Explain why.
-
-
- (b) Where are series circuits used in homes? Explain why.
-
-

3.6.4.11

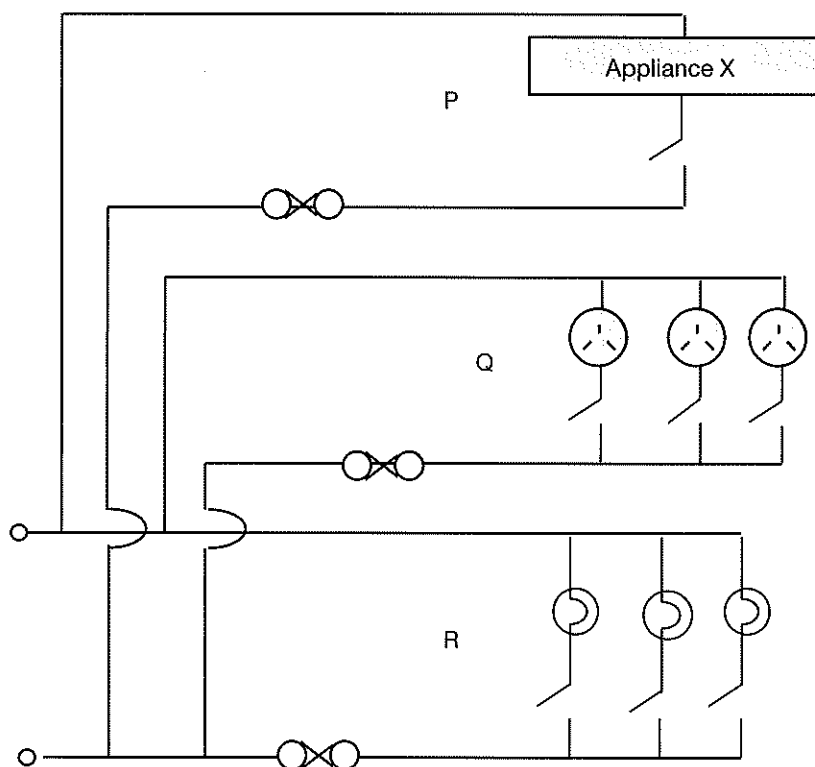
- (a) What are the three different types of circuits used in homes?
-
- (b) Briefly explain why these three types of circuits are used.
-
-
- (c) List the circuits used in homes in order of decreasing amount of maximum current each can carry.
-
- (d) How does the maximum current each circuit can carry affect the wire used to construct the circuit?
-

3.6.4.12 The diagram shows a simplified illustration of the three different types of electrical circuits used in homes.

- (a) Identify the type of circuits labelled P, Q and R.
-

- (b) How are the components connected into each of these three circuits?
-
-

- (c) Suggest three devices appliance X could be.
-
-



3.6.4.13 The diagram shows a simple lighting circuit for a small home.

(a) In which room will one switch turn on more than one light?

(b) Explain the connection of the globes and the switches in room S.

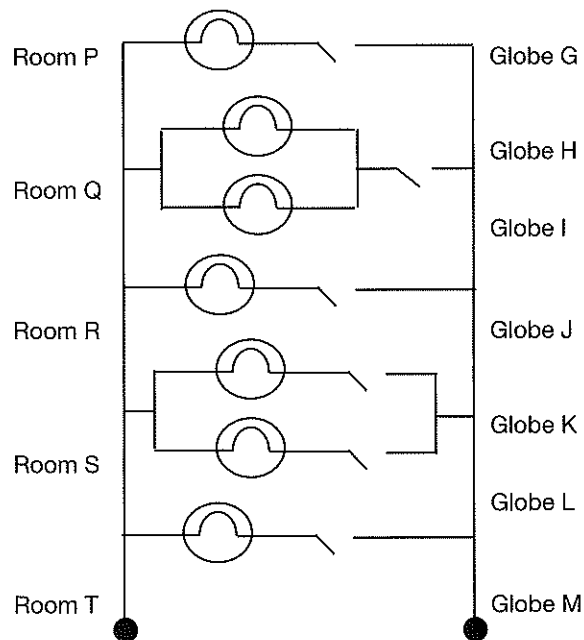
(c) If one of the lights in room Q blows, what happens to the other globe? Justify your answer.

(d) Identify the type of electrical connection used for the lights in a room if one switch turns them all on, but at the same time, if one blows, all the others stay on.

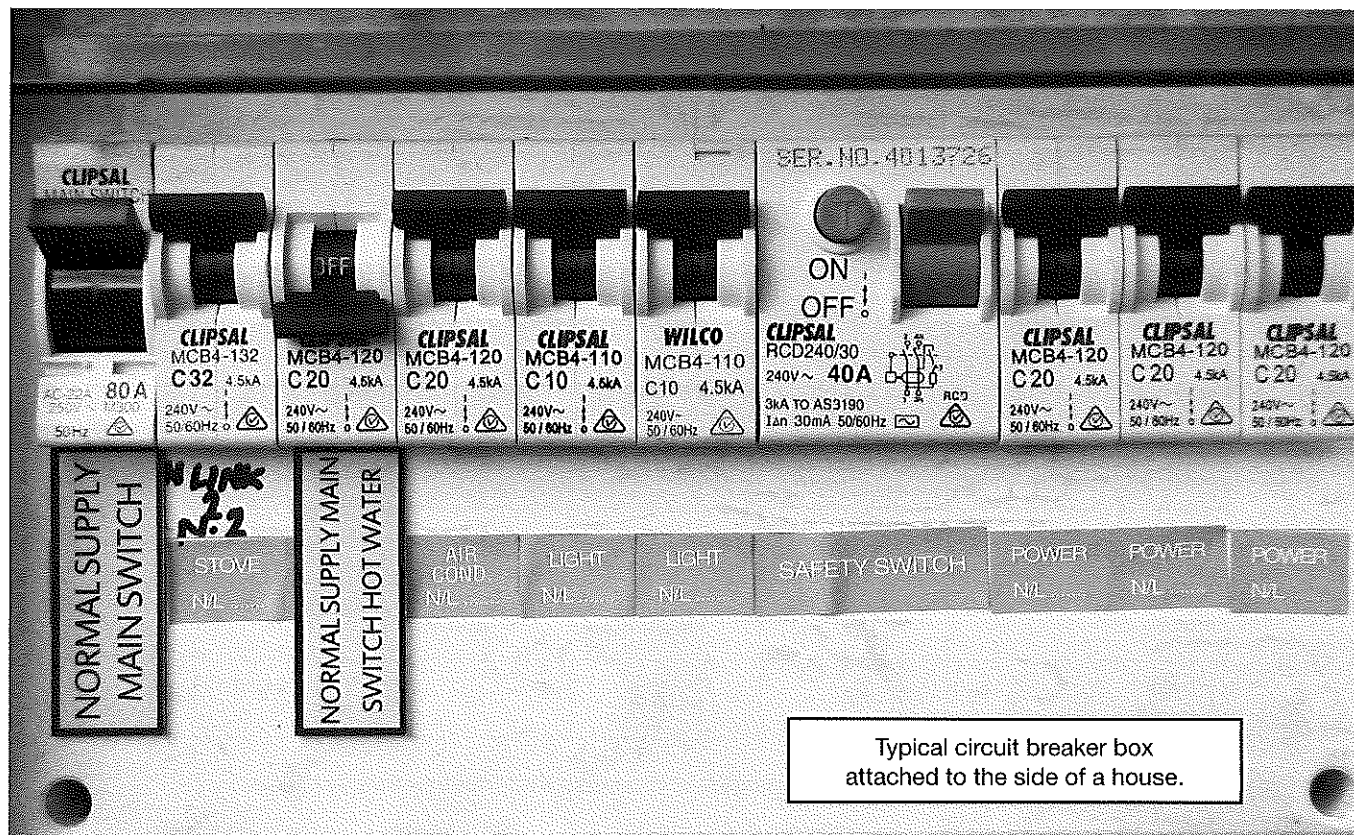
(e) In the space provided, draw a sketch of this circuit to show what you mean. Include a DC power source in your circuit.

(f) If there are four lights in a room and each can be turned on separately, what is the minimum number of switches required?

(g) In the space provided, draw a diagram of the circuit needed for (f).

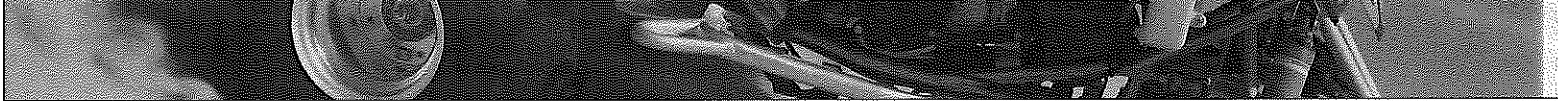


3.6.4.14 The photograph shows the circuit breaker safety switches for a typical household electrical supply.



Typical circuit breaker box attached to the side of a house.

- (a) How many different lighting circuits does this home have?
-
- (b) How many different power circuits does this home have?
-
- (c) How many different dedicated circuits does this home have?
-
- (d) Identify the electrical devices connected into dedicated circuits in this home.
-
- (e) Each safety switch carries a label, for example, C20 or C10. What might these labels mean?
-
- (f) Which of the dedicated circuit appliances allows the largest maximum current? What is the magnitude of this current?
-
- (g) Suggest the maximum current this power board allows into the household circuits. Justify your answer.
-
- (h) The normal supply hot water system switch is off. In fact, it is permanently off in this home. Propose a reason for this.
-
- (i) The safety switch has a label '40 A' on it. What does this mean?
-



3.7 Development of electrical technologies.

3.7.1 Energy efficiency of light sources.

3.7.1.1 Consider the information in the following table.

Type of globe	Incandescent	Fluorescent	Compact fluorescent	LED	Halogen
Average life (hours)	750 to 1000	20 000	10 000	45 000	3000
Watts	3 to 500	4 to 215	3 to 120	2.5 to 16	5 to 500
Lumens per watt	15	60 to 100	60	45	25
Cost to operate	High	Low	Low	Low	Medium
Price	Low	Medium	Medium	High	Medium
Power to produce same light (W)	60	15	13	10	36

Compare the characteristics of these globes and assess their relative efficiencies.

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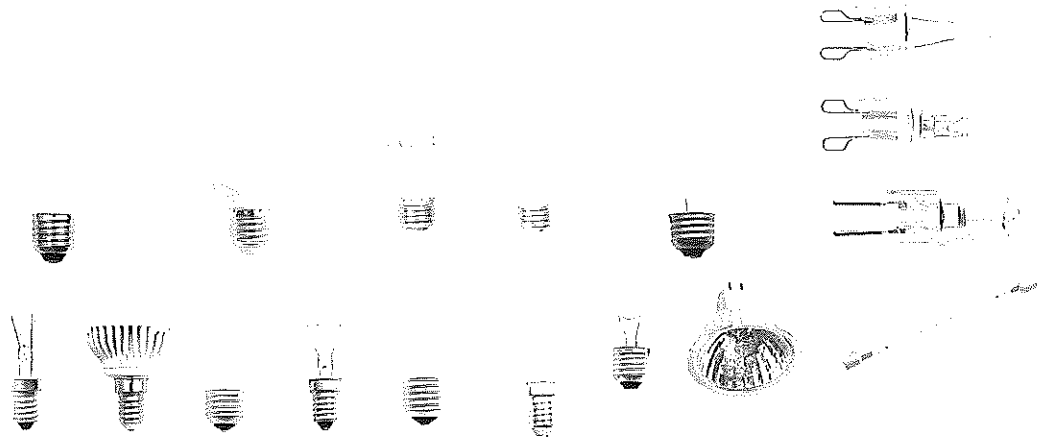
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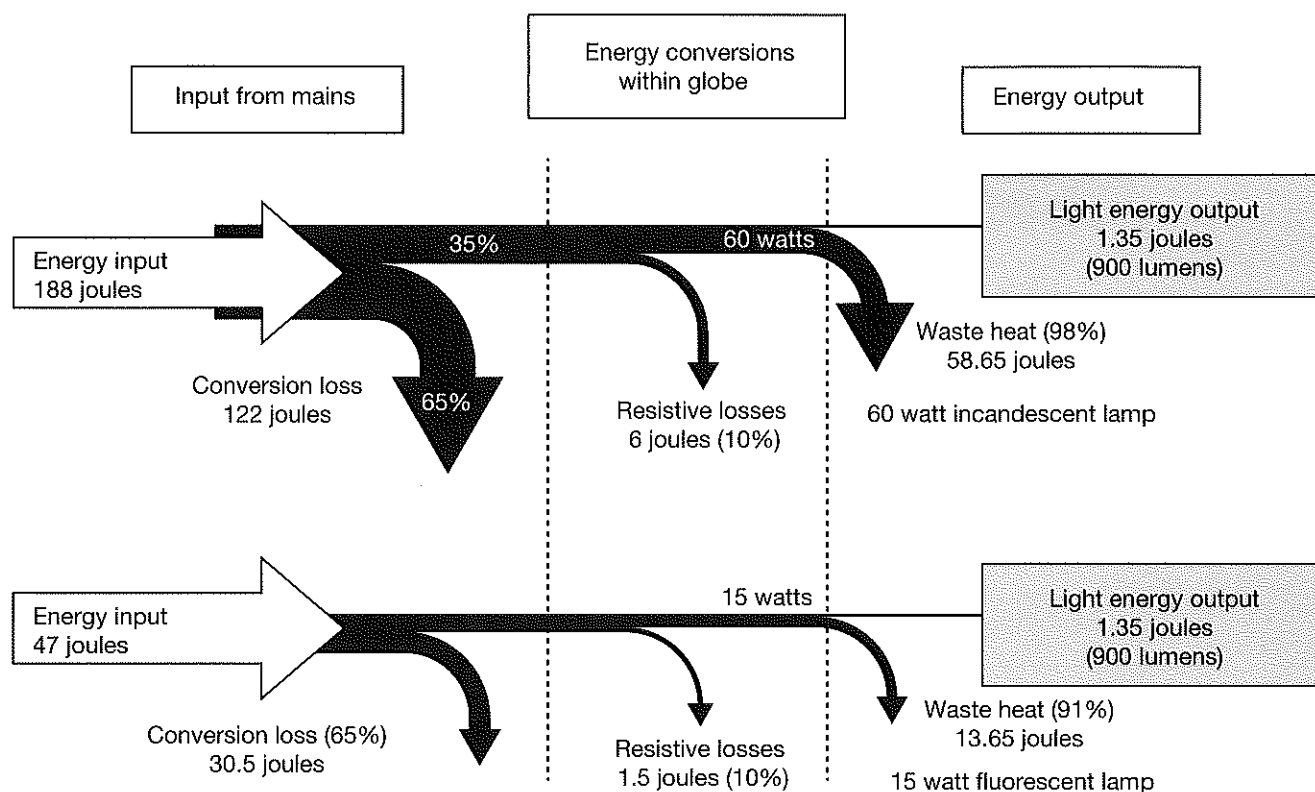
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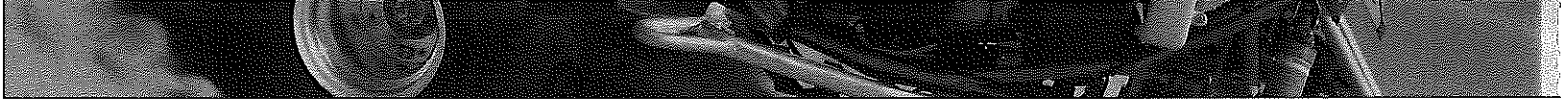
.....



3.7.1.2 Consider the following diagram.



- (a) Explain what this diagram is showing.
- (b) Based on this information, comment on the relative efficiency of the incandescent and fluorescent lamps.
- (c) Assess the following statement: 'The percentage losses of energy by these lamps during conversion of electricity to light, from electrical resistance and by heating effects are all identical. The two lamps must therefore be equally efficient.'
- (d) Some states in Australia are passing regulations to phase out the use of incandescent lamps. Suggest reasons for this.



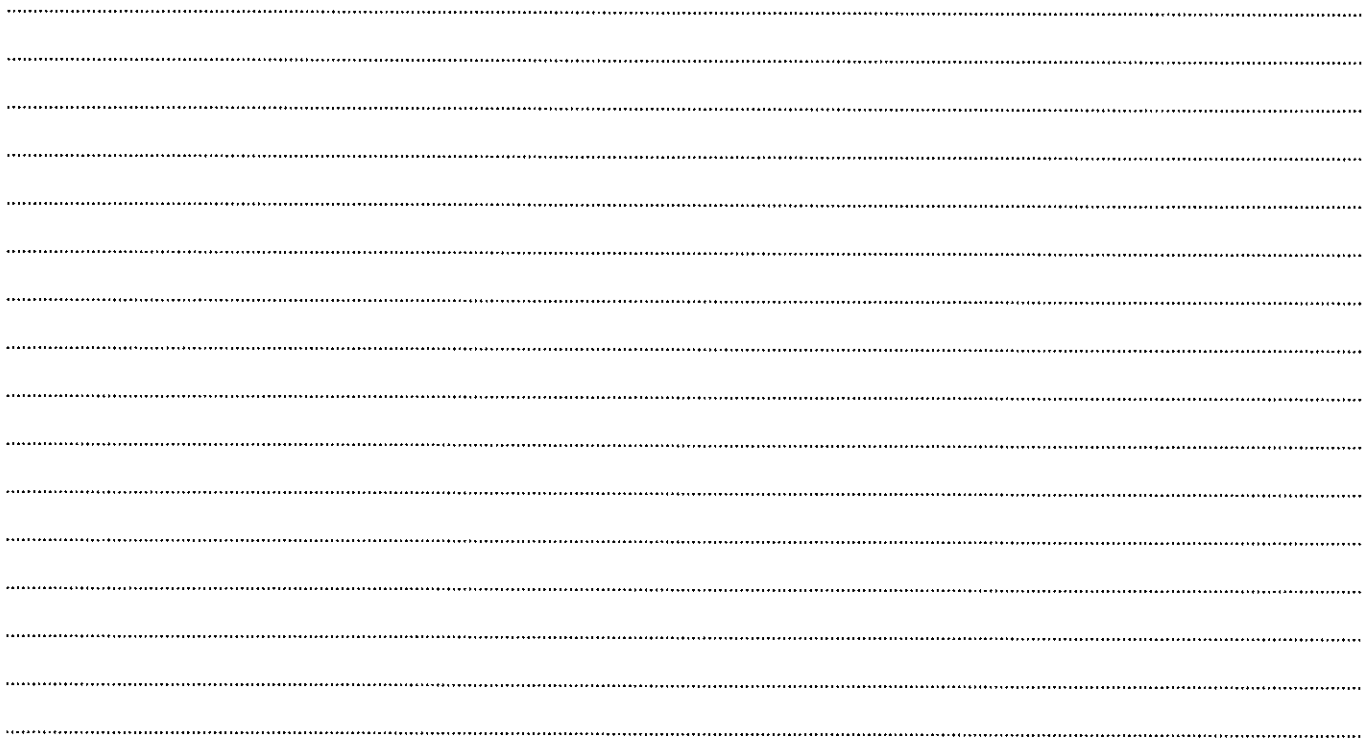
3.7.2 Impacts of electricity.

3.7.2.1 Complete the table by identifying 10 different devices which run on electricity or uses of electricity and then briefly describe the way you think your life would be different if we did not have a regular source of electricity in our community.

	Device running on electricity or use of electricity	Impacts this has had on your life
(a)		
(b)		
(c)		
(d)		
(e)		
(f)		
(g)		
(h)		
(i)		
(j)		

Imagine your life using candles at night, cooking on a wood fired oven, chopping wood for cooking and heating, no radio or TV, no heavy machinery to make roads, no workshop tools.

Word process your answer and summarise your main points here.



3.8 The use of scientific knowledge.

3.8.1 Edison versus Westinghouse.

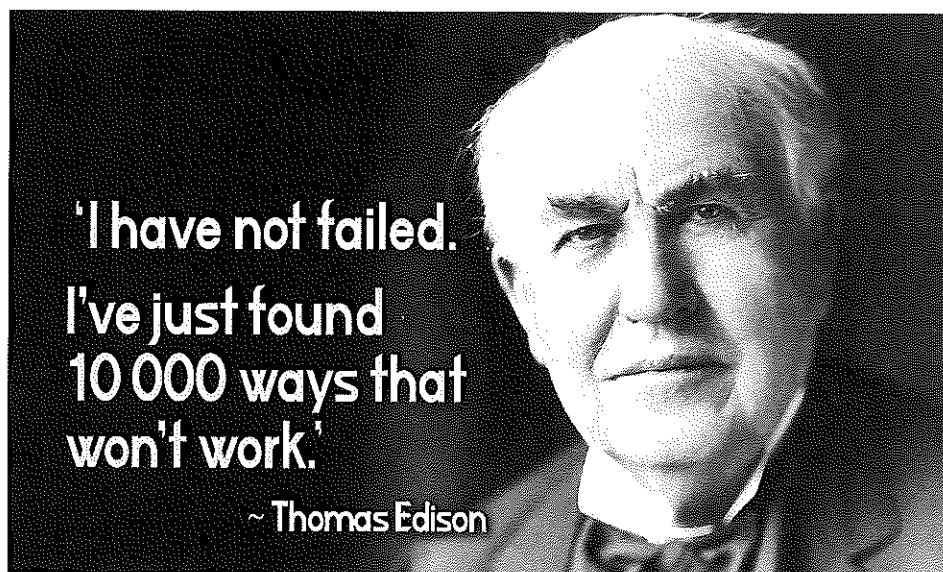
The progress of science throughout time has not always been exciting and friendly. The moral and economic disagreements between Thomas Edison, incorrectly credited with the invention of the light globe and George Westinghouse over the use of DC or AC to provide electricity to people clearly illustrates this. Edison had several cities powered by DC – a lucrative business for him. Although he had no way at the time of generating it, Westinghouse openly proposed that his proposed AC would be better and more efficient.

The summary of events below summarises their disagreement, although it perhaps is more sour grapes from Edison against Westinghouse simply working to develop an alternate electrical energy supply system. Read the information, researching for additional data if you wish, then answer the questions which follow it.

Edison fails with electric light globes

It would seem that Thomas Edison, long credited with the invention of the light globe, copied and patented designs developed at least 10 years earlier by Joseph Swan. Swan's appeal to the British courts for patent infringement was upheld, and Edison was forced to name Swan a partner in his electrical company.

Edison lost a similar battle in America where his patents on the globe were ruled invalid, a judgement based on William Sawyer's earlier work.



The final blow to Edison's ventures with light globes came when Swan sold his patents to the US Brush Electric Company. Edison no longer had access to the patents and could no longer manufacture light globes.

'I have not failed. I've just found 10,000 ways that won't work.' Thomas Edison

Edison distributes DC electricity

Edison changed tack and went into business setting up a direct current (DC) system of distributing electrical power in New York City. Because DC cannot be transformed, its transmission distance is severely limited. Edison had hundreds of generators all over New York, each about one kilometre from the next.

Westinghouse develops AC generators

At the same time (1886) George Westinghouse set up an AC generating system, invented by Nikola Tesla, in competition with Edison's DC, applying Tesla's transformers to transmit at high voltage before stepping it down for household use. Despite buying the European rights to transformers, Edison seems to have mistrusted AC. He continued to agitate for the adoption of DC.

Edison plays dirty?

Edison's tactics were often less than honest. He lobbied politicians to pass laws prohibiting the use of AC above 250 V, while recommending 800 V for his DC. He conducted highly publicised experiments electrocuting hundreds of animals with AC – proving it was unsafe.

Edison's support for an AC electric chair backfires

When New York called for designs for an electric chair to electrocute criminals, Edison, in a further attempt to make AC unpopular with people, recommended the adoption of the unsafe AC, and suggested the name 'Westinghouse Chair' in which criminals would be 'Westinghoused'. Unfortunately for Edison, the first attempt at electrocution was done very badly, the current having to be turned on and off several times. The executed man's body was partially burned, and was too hot to be removed from the chair for some hours. Edison was the person who lost credibility because of his vocal support for the use of the AC.

Westinghouse wins the World Fair contract

Shortly after this, Westinghouse won the bid for illuminating The Chicago World's Fair incorporating the 400th anniversary of the discovery of America by Christopher Columbus, undercutting Edison's \$1 000 000 bid by half.

On 1 May 1893, President Cleveland pushed a button, and a hundred thousand lights came on, powered by twelve AC generators. The 27 000 000 people who visited the fair were suitably impressed, and from then on, over 80% of all new electrical appliances were built for AC.

Westinghouse produces hydroelectricity from Niagara Falls

The final blow to DC distribution came when Westinghouse harnessed the water power from Niagara falls as AC electricity, again defeating Edison's rival proposition. Commissioned in 1891, the first generator built during the five year project provided power for the Buffalo street railway company, over 32 kilometres away. By 1896, ten generators supplied New York City with lights, running street railways and the subway system. Even Edison was convinced of the superiority of AC distribution.



George Westinghouse.

3.8.1.1 Answer this question by discussing your thoughts with your classmates.

Thomas Edison was a great scientist. He had over 150 inventions and held over 1900 scientific patents, yet his tactics in the competition against Westinghouse seem to be a bit unethical. His motives were perhaps threefold – to enhance his reputation as a leading scientist, to profit from the vast amounts of money that would likely flow to the successful combatant, and to protect his substantial scientific reputation.

- Critically evaluate each of Edison's possible motives.
- Assess the fairness of Edison's tactics.
- With so much scientific prestige behind him, was Edison justified to attempt to 'win at all costs'?

3.8.1.2 Westinghouse created opportunities for his AC distribution system at the right time by underbidding Edison on two extremely important projects.

(a) Identify these two projects.

(b) Outline why these two projects were so important to AC distribution.

(c) Imagine that Edison had been successful in winning one or even both of these projects. Predict how this might have affected the final outcome of the AC/DC transmission debate. Give reasons for your prediction.

(d) Outline ways in which household electricity supplies would be different if DC transmission was the way our electricity was supplied.

3.8.1.3

(a) Identify the major limitation in a DC transmission system.

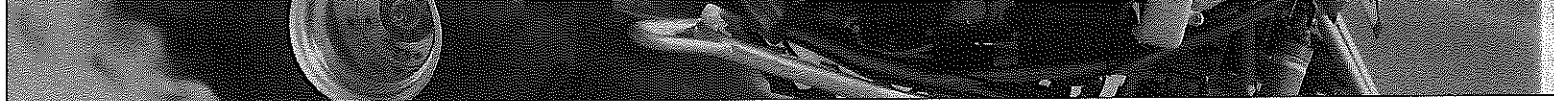
(b) Outline the effects of this limitation on an electrical distribution system to a whole city.

3.8.2 Energy efficiency of batteries.

3.8.2.1

(a) Describe the difference between a battery and a cell.

(b) Compare a dry cell and a wet cell.



3.8.2.2 The table shows the most common types of 'batteries' we use. Complete the table by ranking these batteries in order from least efficient to most efficient.

3.8.2.3 Some factors to be considered when buying a battery, apart from cost, are memory effect, discharge rate and energy density. Explain each of these and how each contributes to the efficiency of a battery.

Type of battery
Alkaline cell
Carbon-zinc
Lead-acid
Lithium ion
Nickel cadmium
Nickel metal hydride

Type of battery	
	Least efficient
	Most efficient

(a) Memory effect.

(b) Discharge rate.

(c) Energy density.

3.8.2.4 Consider the information in the table.

Battery	Voltage (V)	Energy density (W kg ⁻¹)	Number of recharges	Self discharge (%/month)	Memory effect?	Polluting when discarded	Energy cost (W \$ ⁻¹)
X	1.2	40 to 60	Up to 1500	20 to 30	Yes	Yes	1.25 to 2.5
Y	1.2	30 to 80	500 to 1000	30 to 35	No	Yes	2.75
Z	3.6	150 to 200	1200 to 10 000	5 to 10	No	No	2.8 to 5.0

(a) Which of the properties listed in the table most affects the useful life of the battery?

(b) List the three types of batteries in the table in order from longest to shortest 'shelf life'.

(c) Although referred to as batteries, only one in the table is a battery. What are the others? Justify your answer.

(d) Based on these properties, which would be the least efficient battery to buy? Justify your answer.

(e) Which would be the best to use in a laptop or tablet? Justify your answer.